eRAN

Inter-RAT Mobility Load Balancing Feature Parameter Description

Issue 02

Date 2015-03-30

HUAWEI TECHNOLOGIES CO., LTD.

Copyright © Huawei Technologies Co., Ltd. 2016. All rights reserved.

No part of this document may be reproduced or transmitted in any form or by any means without prior written consent of Huawei Technologies Co., Ltd.

Trademarks and Permissions

HUAWEI and other Huawei trademarks are trademarks of Huawei Technologies Co., Ltd.
All other trademarks and trade names mentioned in this document are the property of their respective holders.

Notice

The purchased products, services and features are stipulated by the contract made between Huawei and the customer. All or part of the products, services and features described in this document may not be within the purchase scope or the usage scope. Unless otherwise specified in the contract, all statements, information, and recommendations in this document are provided "AS IS" without warranties, guarantees or representations of any kind, either express or implied.

The information in this document is subject to change without notice. Every effort has been made in the preparation of this document to ensure accuracy of the contents, but all statements, information, and recommendations in this document do not constitute a warranty of any kind, express or implied.

Huawei Technologies Co., Ltd.

Address: Huawei Industrial Base
Bantian, Longgang
Shenzhen 518129
People's Republic of China

Website: http://www.huawei.com

Email: support@huawei.com

Contents

1 About This Document 1
1.1 Scope 1
1.2 Intended Audience 1
1.3 Change History 2
1.4 Differences Between eNodeB Types 2
2 Overview 4
2.1 Introduction 4
2.2 Benefits 4
2.3 Architecture 4
3 Related Concepts 6
3.1 Cell Load 6
3.2 Source Cell and Target Cell 6
3.3 Overlapping Neighboring Cell 6
3.4 Type of UEs to Be Transferred 6
4 MLB to UTRAN 8
4.1 MLB Triggering and Stopping 9
4.2 Target Cell/Frequency Selection 10
4.2.1 Load Information Exchange 10
4.2.2 Target Cell/Frequency Determination 10
4.3 UE Selection 11
4.4 Load Transfer 12
4.4.1 UL-Sync UEs 12
4.4.2 RRC_REL UEs 13
5 MLB to GERAN 15
5.1 MLB Triggering and Stopping 16
5.2 Load Information Exchange 16
5.3 Target Cell Selection 16
5.4 UE Selection and Load Transfer 16
6 Related Features 17
6.1 Features Related to LOFD-001044 Inter-RAT Load Sharing to UTRAN 17
6.2 Features Related to LOFD-001045 Inter-RAT Load Sharing to GERAN 17

Copyright © Huawei Technologies Co., Ltd.

7 Network Impact....18
7.1 LOFD-001044 Inter-RAT Load Sharing to UTRAN....18
7.2 LOFD-001045 Inter-RAT Load Sharing to GERAN....18
8 Engineering Guidelines....20
8.1 LOFD-001044 Inter-RAT Load Sharing to UTRAN....20
8.1.1 When to Use LOFD-001044....20
8.1.2 Required Information....20
8.1.3 Planning....21
8.1.3.1 RF Planning....21
8.1.3.2 Network Planning....21
8.1.3.3 Hardware Planning....21
8.1.4 Requirements....21
8.1.5 Data Preparation....21
8.1.6 Activation....30
8.1.7 Activation Observation....33
8.1.8 Deactivation....37
8.1.9 Reconfiguration....39
8.1.10 Performance Monitoring....39
8.1.11 Parameter Optimization....40
8.1.12 Troubleshooting....40
8.2 LOFD-001045 Inter-RAT Load Sharing to GERAN....40
8.2.1 When to Use LOFD-001045....40
8.2.2 Required Information....40
8.2.3 Planning....41
8.2.3.1 RF Planning....41
8.2.3.2 Network Planning....41
8.2.3.3 Hardware Planning....41
8.2.4 Requirements....41
8.2.5 Data Preparation....41
8.2.6 Activation....45
8.2.7 Activation Observation....48
8.2.8 Deactivation....49
8.2.9 Performance Monitoring....49
8.2.10 Parameter Optimization....50
8.2.11 Troubleshooting....50
9 Parameters....51
10 Counters....72
11 Glossary....76
12 Reference Documents....77

Copyright © Huawei Technologies Co., Ltd.

1 About This Document

1.1 Scope

This document describes inter-radio access technology mobility load balancing (inter-RAT MLB), including its technical principles, related features, network impact, and engineering guidelines.

This document covers the following features:

- LOFD-001044 Inter-RAT Load Sharing to UTRAN
- LOFD-001045 Inter-RAT Load Sharing to GERAN

This document applies to the following types of eNodeBs.

eNodeB Type	Model
Macro	3900 series eNodeB
Micro	BTS3202E and BTS3203E
LampSite	DBS3900

Any managed objects (MOs), parameters, alarms, or counters described herein correspond to the software release delivered with this document. Any future updates will be described in the product documentation delivered with future software releases.

This document applies only to LTE FDD. Any "LTE" in this document refers to LTE FDD, and "eNodeB" refers to LTE FDD eNodeB.

1.2 Intended Audience

This document is intended for personnel who:

- Need to understand the features described herein
- Work with Huawei products

Copyright © Huawei Technologies Co., Ltd.

1.3 Change History

This section provides information about the changes in different document versions. There are two types of changes:

- Feature change

  Changes in features and parameters of a specified version as well as the affected entities
- Editorial change

  Changes in wording or addition of information and any related parameters affected by editorial changes. Editorial change does not specify the affected entities.

eRAN7.0 02 (2015-03-30)

This issue includes the following changes.

Change Type	Change Description	Parameter Change
Feature change	None	None
Editorial change	Revised descriptions in this document.	None

eRAN7.0 01 (2014-06-25)

This document is split from *Mobility Load Balancing Feature Parameter Description* for Issue 02 (2014-05-30) of eRAN7.0.

1.4 Differences Between eNodeB Types

Feature Support by Macro, Micro, and LampSite eNodeBs

Feature ID	Feature Name	Supported by Macro eNodeBs	Supported by Micro eNodeBs	Supported by LampSite eNodeBs
LOFD-001044	Inter-RAT Load Sharing to UTRAN	Yes	Yes	Yes
LOFD-001045	Inter-RAT Load Sharing to GERAN	Yes	No	Yes

Copyright © Huawei Technologies Co., Ltd.

Function Implementation in Macro, Micro, and LampSite eNodeBs

Function	Difference
Carrier aggregation (CA)	Micro eNodeBs do not support CA. For details, see **4.4.1 UL-Sync UEs**.

Copyright © Huawei Technologies Co., Ltd.

2 Overview

2.1 Introduction

Inter-RAT MLB coordinates load distribution among inter-RAT cells. For this purpose, MLB checks the load status of cells and transfers UEs from heavily loaded cells to lightly loaded inter-RAT cells.

2.2 Benefits

Inter-RAT MLB offers the following benefits:

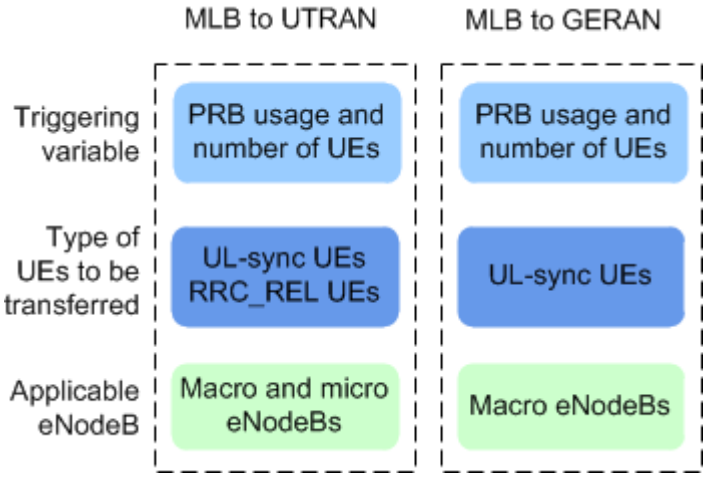
- Relieves load imbalances among inter-RAT cells by transferring appropriate UEs to inter-RAT neighboring cells.
- Increases the access success rate, improves user experience with telecommunication services, and achieves better overall resource utilization.

2.3 Architecture

Inter-RAT MLB can be categorized by triggering variables and the types of UEs to be transferred, as shown in **Figure 2-1**.

Copyright © Huawei Technologies Co., Ltd.

Figure 2-1 Triggering variables and types of UEs to be transferred for inter-RAT MLB



Currently, operator-level inter-RAT MLB does not work in RAN sharing scenarios.

Copyright © Huawei Technologies Co., Ltd.

3 Related Concepts

3.1 Cell Load

eNodeBs determine whether to trigger MLB based on the air interface load. The air interface load is represented by the physical resource block (PRB) usage and the number of uplink-synchronized UEs in a cell.

NOTE

For details about how to calculate PRB usage, see section 4.1.1 "PRB usage" in 3GPP TS 36.314 V10.2.0.

3.2 Source Cell and Target Cell

A source cell is the cell from which MLB transfers the load. In this document, a source cell is also referred to as a serving cell. A target cell is a neighboring cell to which MLB attempts to transfer the load.

3.3 Overlapping Neighboring Cell

In this document, an overlapping neighboring cell is a neighboring UTRAN cell for which the **UtranNCell**.***OverlapInd*** parameter is set to **YES(YES)**. You are advised to set this parameter to **YES(YES)** for neighboring UTRAN cells whose coverage area is the same as or overlaps a large proportion of the serving cell's coverage area.

3.4 Type of UEs to Be Transferred

Inter-RAT MLB transfers uplink-synchronized (UL-sync) UEs and RRC_REL UEs.

- UL-sync UE: If an RRC connection has been established between a UE and an eNodeB and the UE maintains uplink synchronization with the eNodeB, this UE is a UL-sync UE. The eNodeB allocates physical uplink control channel (PUCCH) and sounding reference signal (SRS) resources to the UE.

Copyright © Huawei Technologies Co., Ltd.

- RRC_REL UE: An RRC_REL UE is a UE that an eNodeB is about to release because the UE inactivity timer specified by the **RrcConnStateTimer**.***UeInactiveTimer*** parameter expires.

Transfer of RRC_REL UEs reduces the potential load in terms of the number of UEs in the source cell without affecting user experience. The two transfer methods can be used either separately or collaboratively.

If the eNodeB configures a secondary serving cell (SCell) for a CA UE, the UE is in CA state.

Copyright © Huawei Technologies Co., Ltd.

4 MLB to UTRAN

This chapter describes the feature LOFD-001044 Inter-RAT Load Sharing to UTRAN.

This feature transfers UL-sync UEs or RRC_REL UEs depending on parameter settings:

- MLB to UTRAN transfers UL-sync UEs if both of the following options are selected:
  - **UtranMlbSwitch(UtranMlbSwitch)** option of the **CellAlgoSwitch**.***MlbAlgoSwitch*** parameter
  - **UtranPsHoSwitch(UtranPsHoSwitch)** option of the **EnodebAlgoSwitch**.***HoModeSwitch*** parameter
- MLB to UTRAN transfers RRC_REL UEs if the **UtranIdleMlbSwitch(UtranIdleMlbSwitch)** option of the **CellAlgoSwitch**.***MlbAlgoSwitch*** parameter is selected.

If both MLB to UTRAN and MLB to GERAN are enabled, the two procedures work in parallel. For details about MLB to GERAN, see **5 MLB to GERAN**.

Copyright © Huawei Technologies Co., Ltd.

4.1 MLB Triggering and Stopping

After MLB to UTRAN is enabled, an eNodeB periodically collects the PRB usage and the number of UL-sync UEs in each cell.

MLB is triggered or stopped when any of the following types of PRB usage meets certain conditions:

- Uplink guaranteed bit rate (GBR) PRB usage
- Downlink GBR PRB usage
- Uplink total PRB usage (uplink GBR + non-GBR)
- Downlink total PRB usage (downlink GBR + non-GBR)

GBR PRB usage takes priority over total PRB usage in triggering and stopping MLB. If the GBR PRB usage meets the specific triggering condition, MLB is triggered by GBR PRB usage and the eNodeB will transfer GBR services. If the GBR PRB usage does not meet the triggering condition but the total PRB usage does, MLB is triggered by total PRB usage and the eNodeB will transfer non-GBR services.

MLB by Transferring UL-Sync UEs

The eNodeB triggers a procedure for "MLB to UTRAN by transferring UL-sync UEs" if a cell meets both of the following conditions throughout 5s:

- PRB usage of the cell ≥ **CellMLB**.***InterRatMlbThd*** + **CellMLB**.***LoadOffset***
- Number of UL-sync UEs in the cell ≥ **CellMLB**.***InterRatMlbUeNumThd*** + **CellMLB**.***InterRatMlbUeNumOffset***

The eNodeB stops the MLB procedure if the cell meets one of the following conditions throughout 5s:

- PRB usage of the cell < **CellMLB**.***InterRatMlbThd***
- Number of UL-sync UEs in the cell < **CellMLB**.***InterRatMlbUeNumThd***

MLB by Transferring RRC_REL UEs

The eNodeB triggers a procedure for "MLB to UTRAN by transferring RRC_REL UEs" if a cell meets both of the following conditions throughout 5s:

- PRB usage of the cell ≥ **CellMLB**.***InterRatMlbThd*** + **CellMLB**.***LoadOffset***
- Number of UL-sync UEs in the cell ≥ **CellMLB**.***InterRatIdleMlbUeNumThd*** + **CellMLB**.***InterRatMlbUeNumOffset***

The eNodeB stops the MLB procedure if the cell meets one of the following conditions throughout 5s:

- PRB usage of the cell < **CellMLB**.***InterRatMlbThd***
- Number of UL-sync UEs in the cell < **CellMLB**.***InterRatIdleMlbUeNumThd***

Copyright © Huawei Technologies Co., Ltd.

4.2 Target Cell/Frequency Selection

After MLB to UTRAN is triggered, the eNodeB performs target cell selection within each MLB period specified by the **CellMLB**.***InterRatLoadEvalPrd*** parameter. If the target cell list is empty in a period, the eNodeB does not execute MLB in this period.

4.2.1 Load Information Exchange

An eNodeB can obtain the load information of neighboring UTRAN cells through RAN information management (RIM) procedures. This function requires that the UTRAN should support the RIM procedures for load information exchange as defined in 3GPP Release 10.

In load information exchange with UTRAN, the **GlobalProcSwitch**.***UtranLoadTransChan*** parameter must be set to **BASED_ON_RIM** on the LTE network, and the load information exchange function must be enabled in the UTRAN. If the peer RNC is a Huawei device, see *Interoperability Between UMTS and LTE* for details.

If the **MlbLoadInfoSwitch(MlbLoadInfoSwitch)** option of the **CellAlgoSwitch**.***MlbAlgoSwitch*** parameter is selected for an E-UTRAN cell, the cell obtains load information about all neighboring UTRAN cells through the RIM procedures. If the option is not selected, the E-UTRAN cell does not consider the load status of neighboring UTRAN cells.

In RIM procedures, messages can be exchanged through the core network or the eCoordinator, a Huawei-proprietary NE, depending on the setting of the **EnodeBAlgoSwitch**.***RimOnEcoSwitch*** parameter:

- If the **EnodeBAlgoSwitch**.***RimOnEcoSwitch*** parameter is set to **OFF(Off)**, messages are exchanged through the core network. This requires that the core network support the RIM procedures defined in 3GPP Release 9.

  For details about the RIM procedures for load information exchange, see section 8c.2 "RIM procedures" in 3GPP TS 48.018 V10.6.0 and sections 8.13 "eNB Direct Information Transfer" and 8.14 "MME Direct Information Transfer" in 3GPP TS 36.413 V10.6.0.
- If the **EnodeBAlgoSwitch**.***RimOnEcoSwitch*** parameter is set to **ON(On)** and the RIM function is enabled on the eCoordinator, messages are exchanged through the eCoordinator.

  This type of RIM procedure involves the eNodeB, eCoordinator, and RNC. It requires that the RIM function be enabled on all the three NEs.

NOTE

In eCoordinator-based RIM procedures, the preceding NEs neither send RIM messages to the core network nor respond to RIM messages sent from the core network.

For details about RIM procedures between E-UTRAN and UTRAN, see *CS Fallback Feature Parameter Description*.

4.2.2 Target Cell/Frequency Determination

Based on load information, the eNodeB filters out the following cells from the UTRAN neighboring relation table (NRT) to generate a list of candidate cells:

- Congested or overloaded cells

Copyright © Huawei Technologies Co., Ltd.

- Cells for which the **UtranNCell**.***NoHoFlag*** parameter is set to **FORBID_HO_ENUM(Forbid Ho)**

Target Cells for MLB by Transferring UL-Sync UEs

The eNodeB selects target frequencies from the operating frequencies of candidate cells. The cells on the selected frequencies are the target cells to be transferred within the current MLB period. The selection policy is determined by the **CellMLB**.***FreqSelectStrategy*** parameter:

- If the parameter is set to **FAIRSTRATEGY(FairStrategy)**, the eNodeB selects all the operating frequencies of the candidate cells as target frequencies. If inter-frequency MLB and MLB to UTRAN are triggered simultaneously, the two procedures work in parallel without affecting each other.
- If the parameter is set to **PRIORITYBASED(PriorityBased)**, the eNodeB selects the frequencies (one or more frequencies) with the highest priority defined by **UtranNFreq**.***MlbFreqPriority*** as target frequencies. If both inter-frequency MLB and MLB to UTRAN are triggered simultaneously, the eNodeB considers all frequency priorities defined by the **EutranInterNFreq**.***MlbFreqPriority*** and **UtranNFreq**.***MlbFreqPriority*** parameters and selects the highest-priority frequencies as target frequencies.

NOTE

An eNodeB imposes a penalty on a frequency selected based on MLB priorities if no UE is transferred to the frequency within the current MLB period defined by **CellMLB**.***InterRatLoadEvalPrd***. During the penalty, the eNodeB no longer selects the frequency as a target frequency. The penalty starts from the next MLB period and lasts for four MLB periods.

Target Frequencies for MLB by Transferring RRC_REL UEs

After generating a list of candidate cells, the eNodeB selects target frequencies for RRC_REL UEs. A UTRAN frequency can be a target if either of the following conditions is met:

- The neighboring cells on the frequency are overlapping neighboring cells, and the load status of these cells is light or invalid.
- None of the neighboring cells on the frequency are overlapping neighboring cells, and the load status of any of these neighboring cells is light or invalid.

4.3 UE Selection

MLB by Transferring UL-Sync UEs

If both MLB to UTRAN and PS handover to UTRAN are enabled, the eNodeB selects a number of UEs based on information about target cells, PRB usage type that triggers MLB, and PRB usage of UEs in the serving cell.

The eNodeB preferentially selects non-CA UEs over CA UEs. However, if the eNodeB is a micro eNodeB, it does not support CA and therefore does not check the CA capability of UEs.

For UEs configured with subscriber profile IDs (SPIDs), the eNodeB selects UEs for which the **SpidCfg**.***InterRatMlbSwitch*** parameter is set to **TRUE(TRUE)**.

The eNodeB selects UEs based on the policy specified by the **CellMLB**.***InterRatMlbUeSelStrategy*** parameter:

Copyright © Huawei Technologies Co., Ltd.

- If the parameter is set to **PRBRATIOHIGHMODE(PRB Ratio High Mode)**, the PRB usage of UEs is the major contributor to the triggering of MLB. The MLB algorithm imposes constraints on both uplink and downlink PRB usage of UEs to be selected. Each of the UEs must meet all the following conditions:
  - PRB usage of the UE in the direction (uplink or downlink) where MLB is triggered > 2%
  - Total PRB usage of all UEs to be transferred in the same direction < **CellMLB**.***LoadDiffThd***/2
  - PRB usage of the UE in the opposite direction ≤ 2%

  The eNodeB evaluates both uplink and downlink PRB usage of individual UEs to prevent MLB back to the source cell. If the uplink or downlink PRB usage of the selected UEs is too high, there is a high probability of further triggering uplink or downlink MLB from the target cell. If the uplink or downlink PRB usage of the selected UEs is too low, the source cell load may decrease too slowly.
- If this parameter is set to **PRBRATIOLOWMODE(PRB Ratio Low Mode)**, the number of UEs is the major contributor to the triggering of MLB. The MLB algorithm selects UEs with low PRB usage so that more UEs can be transferred to inter-RAT neighboring cells rapidly and the impact on user experience is reduced. To be specific, the eNodeB selects UEs whose PRB usage is less than the **CellMLB**.***InterRatMlbUeSelPrbThd*** parameter value.

MLB by Transferring RRC_REL UEs

The eNodeB selects UEs that meet the following conditions:

- The cause for the RRC connection release is that the timer defined by **RrcConnStateTimer**.***UeInactiveTimer*** for the UE expires.
- The UE supports the target frequencies.
- The serving PLMN set of the UE and the home PLMNs of external cells on the target frequencies have an intersection.
- If the UE is configured with SPID-specific dedicated cell-reselection priorities of frequencies, the frequencies and target frequencies have an intersection.

4.4 Load Transfer

4.4.1 UL-Sync UEs

There are two methods of transferring UEs that support inter-RAT handover: measurement-based handover and blind handover. The eNodeB transfers these UEs from the E-UTRAN cell to the target cells based on inter-RAT handover policies.

Measurement-based Handover

The MLB algorithm requires that the received signal code power (RSCP) of target UTRAN cells be greater than the **InterRatHoUtranGroup**.***LdSvBasedHoUtranB1ThdRscp*** parameter value or the Ec/No of target UTRAN cells be greater than the **InterRatHoUtranGroup**.***LdSvBasedHoUtranB1ThdEcn0*** parameter value.

Copyright © Huawei Technologies Co., Ltd.

Blind Handover

Blind handovers work only if the **InterRatMlbBlindHo(InterRatMlbBlindHo)** option of the **CellAlgoSwitch**.***MlbHoMode*** parameter is selected. If UEs do not support inter-RAT handovers, the eNodeB transfers the UEs by blind redirection.

Implementation Suggestion

Handover is recommended for load transfer, because the use of redirection interrupts services. eNodeBs transfer UEs to UTRAN cells through redirections if PS handover is enabled but the UEs do not support handovers or if PS handover is disabled but redirection is enabled.

For details about inter-RAT handovers, see *Inter-RAT Mobility Management in Connected Mode Feature Parameter Description*.

4.4.2 RRC_REL UEs

If the **UtranIdleMlbSwitch(UtranIdleMlbSwitch)** option of the **CellAlgoSwitch**.***MlbAlgoSwitch*** parameter is selected and the load of the serving cell meets the conditions for triggering MLB to UTRAN, the eNodeB instructs some RRC_REL UEs to preferentially camp on neighboring UTRAN cells.

The actual duration of MLB to UTRAN is calculated based on the initial duration specified by **CellMLB**.***InitValidPeriod*** and the number of UL-sync UEs. A larger value of this parameter or a larger number of UL-sync UEs results in a longer actual duration, within which RRC_REL UEs preferentially camp on neighboring UTRAN cells.

In MLB to UTRAN, the eNodeB delivers the priorities of target frequencies to each target UE as described in the following sections.

Selecting Frequencies

The eNodeB selects the E-UTRA and inter-RAT frequencies supported by the UE. The UTRA frequencies are those selected as described in **Target Frequencies for MLB by Transferring RRC_REL UEs**.

Determining the Cell-Reselection Priority Type

If SPID- or operator-specific dedicated cell-reselection priorities of frequencies are defined for an RRC_REL UE, the eNodeB uses the priorities for the UE. For details, see *Flexible User Steering Feature Parameter Description* and *RAN Sharing Feature Parameter Description*. If neither type of priorities is configured for the UE, the eNodeB uses the absolute priorities of the frequencies.

Determining and Delivering Frequency Priorities

In the IdleModeMobilityControlInfo IE, the eNodeB includes the priorities of the frequencies in descending order of frequency categories:

1. Target UTRA frequency
2. Non-target UTRA frequency
3. E-UTRA frequency
4. GERAN frequency

Copyright © Huawei Technologies Co., Ltd.

If there is more than one frequency in a category, the priorities of these frequencies are dependent on the SPID-specific dedicated priorities, operator-specific dedicated priorities, or absolute priorities for cell reselection.

Copyright © Huawei Technologies Co., Ltd.

5 MLB to GERAN

This chapter describes the feature LOFD-001045 Inter-RAT Load Sharing to GERAN.

This feature transfers UL-sync UEs. It is enabled if both of the following options are selected:

- **GeranMlbSwitch(GeranMlbSwitch)** option of the **CellAlgoSwitch**.***MlbAlgoSwitch*** parameter
- **GeranPsHoSwitch(GeranPsHoSwitch)** option of the **EnodebAlgoSwitch**.***HoModeSwitch*** parameter

If both MLB to UTRAN and MLB to GERAN are enabled, the two procedures work in parallel. For details about MLB to UTRAN, see **4 MLB to UTRAN**.

Copyright © Huawei Technologies Co., Ltd.

5.1 MLB Triggering and Stopping

The mechanisms for triggering and stopping MLB to GERAN are the same as those for triggering and stopping the function of MLB to UTRAN by transferring UL-sync UEs. For details, see **4.1 MLB Triggering and Stopping**.

5.2 Load Information Exchange

MLB to GERAN does not support load information exchange.

5.3 Target Cell Selection

After MLB to GERAN is triggered, the eNodeB selects target cells within each MLB period specified by the **CellMLB**.***InterRatLoadEvalPrd*** parameter. If the target cell list is empty in a period, the eNodeB does not execute MLB in this period.

The eNodeB determines target cells by filtering out neighboring GERAN cells for which the **GeranNcell**.***NoHoFlag*** parameter is set to **FORBID_HO_ENUM(Forbid Ho)**.

5.4 UE Selection and Load Transfer

MLB to GERAN does not support transfer of RRC_REL UEs.

The principles for selecting UEs in MLB to GERAN are the same as those for selecting UL-sync UEs in MLB to UTRAN. For details, see **4.3 UE Selection**.

Load transfer in MLB to GERAN is similar to that in MLB to UTRAN. The difference is that measurement-based handovers for MLB to GERAN require the received signal strength indicator (RSSI) of the target GERAN cells to be greater than the value of the **InterRatHoGeranGroup**.***LdSvBasedHoGeranB1Thd*** parameter.

For details about MLB to UTRAN, see **4.4 Load Transfer**. For details about inter-RAT handovers, see *Inter-RAT Mobility Management in Connected Mode Feature Parameter Description*.

Copyright © Huawei Technologies Co., Ltd.

6 Related Features

6.1 Features Related to LOFD-001044 Inter-RAT Load Sharing to UTRAN

Prerequisite Features

LOFD-001044 Inter-RAT Load Sharing to UTRAN requires LOFD-001019 PS Inter-RAT Mobility between E-UTRAN and UTRAN.

Mutually Exclusive Features

None

Impacted Features

None

6.2 Features Related to LOFD-001045 Inter-RAT Load Sharing to GERAN

Prerequisite Features

LOFD-001045 Inter-RAT Load Sharing to GERAN requires LOFD-001020 PS Inter-RAT Mobility between E-UTRAN and GERAN.

Mutually Exclusive Features

None

Impacted Features

None

Copyright © Huawei Technologies Co., Ltd.

7 Network Impact

7.1 LOFD-001044 Inter-RAT Load Sharing to UTRAN

System Capacity

The Inter-RAT Load Sharing to UTRAN feature transfers partial load from an E-UTRAN cell to neighboring UTRAN cells before the source E-UTRAN cell becomes congested, decreasing the E-UTRAN load and improving the user experience of UEs in the E-UTRAN. However, this feature increases the UTRAN load and negatively affects the user experience of UEs in the UTRAN.

If the RIM function is enabled in the E-UTRAN and UTRAN, the impact of MLB on UE throughput decreases. After MLB to UTRAN is enabled, operators can evaluate feature gains by monitoring eNodeB counters related to PRB usage, XPU board usage changes in Huawei RNCs, changes in the overall throughput of UTRAN cells, and changes in the overall throughput of E-UTRAN cells. Assume that the PRB usage of an E-UTRAN cell is 90%, and the PRB usage threshold for triggering inter-RAT MLB is 83%. After MLB to UTRAN is enabled, the PRB usage of the E-UTRAN cell decreases by 7% at most. The load decrease in the E-UTRAN cell is dependent on the load of UTRAN cells. The load of a UTRAN cell can be evaluated based on the number of times the cell enters the load reshuffling (LDR) state and the duration in which the cell stays in the LDR state. The lower the load of the UTRAN cells, the higher the efficiency of the load decrease in the E-UTRAN cell.

Network Performance

The Inter-RAT Load Sharing to UTRAN feature increases the number of inter-RAT handovers from E-UTRAN to UTRAN and the L.RRC.ConnReq.Att.MoSig counter value.

7.2 LOFD-001045 Inter-RAT Load Sharing to GERAN

System Capacity

The Inter-RAT Load Sharing to GERAN feature transfers partial load from an E-UTRAN cell to neighboring GERAN cells before the source E-UTRAN cell becomes congested, decreasing the E-UTRAN load and improving the user experience of UEs in the E-UTRAN.

Copyright © Huawei Technologies Co., Ltd.

However, this feature increases the GERAN load and negatively affects the user experience of UEs in the GERAN.

Network Performance

The Inter-RAT Load Sharing to GERAN feature increases the number of inter-RAT handovers from E-UTRAN to GERAN and the L.RRC.ConnReq.Att.MoSig counter value.

Copyright © Huawei Technologies Co., Ltd.

8 Engineering Guidelines

8.1 LOFD-001044 Inter-RAT Load Sharing to UTRAN

8.1.1 When to Use LOFD-001044

Use LOFD-001044 Inter-RAT Load Sharing to UTRAN when one of the following conditions is met:

- The serving cell has no inter-frequency neighboring cells but overlapping neighboring UTRAN cells.
- The serving cell has both overlapping neighboring E-UTRAN cells and overlapping neighboring UTRAN cells, and the load of the serving cell is high.
- Multi-mode base stations are used, or base stations of different RATs provide contiguous coverage.
- The PRB usage of E-UTRAN cells is high, and the number of UL-sync UEs in the cells is large.

8.1.2 Required Information

Collect the following information:

- Information about each neighboring cell of the cells served by the local eNodeB
  - Whether information about the neighboring cell is complete
  - Whether the neighboring cell has been blacklisted
  - Whether the **No handover indicator** parameter is set to **PERMIT_HO_ENUM(Permit Ho)**
- Status of the X2 interfaces with neighboring eNodeBs
- UE capabilities

  The proportion of UEs that support UTRAN measurements
- Whether the operator has purchased the required licenses
- PRB usage and number of UL-sync UEs in cells within the area where MLB is to be activated

Copyright © Huawei Technologies Co., Ltd.

8.1.3 Planning

8.1.3.1 RF Planning

The live network must provide contiguous coverage:

- No holes exist in the coverage.
- Overshoot coverage is minimized.
- Pilot pollution is minimized.
- Uplink and downlink imbalances are minimized.

8.1.3.2 Network Planning

N/A

8.1.3.3 Hardware Planning

N/A

8.1.4 Requirements

There are no requirements for the operating system and transmission networking. Before deploying LOFD-001044 Inter-RAT Load Sharing to UTRAN, the operator must purchase and activate the license for the feature listed in the following table.

Feature ID	Feature Name	License Control Item	NE	Sales Unit
LOFD-001044	Inter-RAT Load Sharing to UTRAN	Inter-RAT Load Sharing to UTRAN (FDD)	eNodeB	per RRC Connected User

8.1.5 Data Preparation

This section describes the data that you need to collect for setting parameters. Required data is data that you must collect for all scenarios. Collect scenario-specific data when necessary for a specific feature deployment scenario.

There are three types of data sources:

- Network plan (negotiation not required): parameter values planned and set by the operator
- Network plan (negotiation required): parameter values planned by the operator and negotiated with the EPC or peer transmission equipment
- User-defined: parameter values set by users

Required Data

The following table describes the parameter that must be set in a **CellMLB** MO to control the MLB algorithm for a cell.

Copyright © Huawei Technologies Co., Ltd.

Parameter Name	Parameter ID	Data Source	Setting Notes
Load Offset	**CellMLB**.***LoadOffset***	Network plan (negotiation not required)	This parameter specifies the offset applied to the threshold for triggering MLB. This mechanism helps prevent load fluctuations from frequently triggering and stopping MLB. The recommended value is **8**.

The following table describes the parameter that must be set in a **CellAlgoSwitch** MO to specify the handover policy for MLB from a cell.

Parameter Name	Parameter ID	Data Source	Setting Notes
Load Balancing Handover Policy	**CellAlgoswitch**.***MlbHoMode***	Network plan (negotiation not required)	If the **InterRatMlbBlind-Ho(InterRatMlbBlindHo)** option of this parameter is selected, the eNodeB preferentially uses blind inter-RAT handovers for load transfer to UTRAN or GERAN cells. If this option is not selected, blind inter-RAT handovers are disabled.

The following table describes the parameters that must be set in an **InterRatHoUtranGroup** MO to control inter-RAT handovers from an E-UTRAN cell to UTRAN.

Copyright © Huawei Technologies Co., Ltd.

Parameter Name	Parameter ID	Data Source	Setting Notes
Load Service Based UTRAN EventB1 RSCP trigger threshold	**InterRatHoUtranGroup**.***LdSvBasedHoUtranB1ThdRscp***	Network plan (negotiation not required)	This parameter specifies the RSCP threshold for event B1 related to load- or service-based inter-RAT handovers to UTRAN. Inter-RAT handovers to UTRAN require that the RSCP of target UTRAN cells meet certain signal quality requirements. When the measured RSCP exceeds this threshold, a UE sends a measurement report to the eNodeB. The recommended value is **-101**.
Load Service Based UTRANB1 ECN0 threshold	**InterRatHoUtranGroup**.***LdSvBasedHoUtranB1ThdEcn0***	Network plan (negotiation not required)	This parameter specifies the Ec/No threshold for event B1 related to load- or service-based inter-RAT handovers to UTRAN. Inter-RAT handovers to UTRAN require that the Ec/No of target UTRAN cells meet certain signal quality requirements. When the measured Ec/No exceeds this threshold, a UE sends a measurement report to the eNodeB. The recommended value is **-18**.

The following table describes the parameter that must be set in a **UtranNCell** MO to configure the neighbor relationship of an E-UTRAN cell with a UTRAN cell.

Copyright © Huawei Technologies Co., Ltd.

Parameter Name	Parameter ID	Data Source	Setting Notes
Overlap Indicator	**UtranNCell**.***OverlapInd***	Network plan (negotiation not required)	This parameter specifies whether to manually identify the UTRAN cell as an overlapping neighboring cell. The default value is **NO(No)**. If this parameter is set to **YES(Yes)**, the UTRAN cell is manually identified as an overlapping neighboring cell. If this parameter is set to **NO(No)**, the UTRAN cell is not manually identified as an overlapping neighboring cell.

Scenario-specific Data

The following table describes the parameter that must be set in a **CellAlgoSwitch** MO to enable MLB to UTRAN.

Parameter Name	Parameter ID	Data Source	Setting Notes
Load balancing algorithm switch	**CellAlgoSwitch**.***MlbAlgoSwitch***	Network plan (negotiation not required)	The **UtranMlbSwitch(UtranMlbSwitch)** option of this parameter specifies whether to enable MLB to UTRAN. This algorithm is enabled only if this option is selected.
		Network plan (negotiation not required)	The **UtranIdleMlbSwitch(UtranIdleMlbSwitch)** option of this parameter specifies whether to enable the algorithm of "MLB to UTRAN by transferring RRC_REL UEs". This algorithm is enabled only if this option is selected.

Copyright © Huawei Technologies Co., Ltd.

Parameter Name	Parameter ID	Data Source	Setting Notes
		Network plan (negotiation not required)	The **MlbLoadInfoSwitch(MlbLoadInfoSwitch)** option of this parameter specifies whether the eNodeB considers the load status of neighboring UTRAN cells when evaluating MLB to UTRAN. The eNodeB considers the load status of neighboring UTRAN cells only if this option is selected.

The following table describes the parameter that must be set in the **ENodeBAlgoSwitch** MO to enable eCoordinator-based load information exchange.

Parameter Name	Parameter ID	Data Source	Setting Notes
Support RIM by eCoordinator Switch	**ENodeBAlgoSwitch**.***RimOnEcoSwitch***	Network plan (negotiation not required)	If this parameter is set to **ON(On)**, the eNodeB obtains UTRAN load information through an eCoordinator in RIM procedures. If this parameter is set to **OFF(Off)**, the eNodeB obtains UTRAN load information through the core network in RIM procedures.

The following table describes the parameter that must be set in the **ENodeBAlgoSwitch** MO to specify handover policies for MLB to UTRAN.

Copyright © Huawei Technologies Co., Ltd.

Parameter Name	Parameter ID	Data Source	Setting Notes
Handover Mode switch	**ENodeBAlgoSwitch**.***HoModeSwitch***	Network plan (negotiation not required)	If the **UtranPsHoSwitch(UtranPsHoSwitch)** option of this parameter is selected, the eNodeB can transfer UEs to neighboring UTRAN cells through handovers. If this option is not selected, handovers to UTRAN are disabled.

The following table describes the parameters that must be set in a **CellMLB** MO to configure the inter-RAT MLB algorithm for a cell.

Parameter Name	Parameter ID	Data Source	Setting Notes
Inter-RAT Mobility Load Balancing Threshold	**CellMLB**.***InterRatMlbThd***	Network plan (negotiation not required)	This parameter specifies the PRB usage threshold for inter-RAT MLB. The default value is **75**.
Inter-RAT Idle MLB UE Number Threshold	**CellMLB**.***InterRatIdleMlbUeNumThd***	Network plan (negotiation not required)	This parameter specifies the threshold for the number of UL-sync UEs in a cell used to evaluate triggering of "MLB to UTRAN by transferring RRC_REL UEs". If this parameter is set to a large value, there is a low probability of triggering MLB, resulting in low UE throughput. If this parameter is set to a small value, the UE throughput decreases after inter-RAT cell reselection.

Copyright © Huawei Technologies Co., Ltd.

Parameter Name	Parameter ID	Data Source	Setting Notes
Inter-RAT Mlb UE Number Offset	**CellMLB**.***InterRatMlbUeNumOffset***	Network plan (negotiation not required)	This parameter specifies the offset relative to the number of UL-sync UEs used to evaluate triggering of inter-RAT MLB. A small value of this parameter cannot reduce the probability of ping-pong MLB.
UTRAN Idle Mode Mobility Load Balancing Initial Valid Period	**CellMLB**.***InitValidPeriod***	Network plan (negotiation not required)	This parameter specifies the initial duration for MLB to UTRAN by transferring RRC_REL UEs. A larger value of this parameter results in a larger number of RRC_REL UEs transferred to UTRAN upon an overload in the E-UTRAN cell. The recommended value is **10**.

Copyright © Huawei Technologies Co., Ltd.

Parameter Name	Parameter ID	Data Source	Setting Notes
Inter-RAT Mobility Load Balancing UE Number Threshold	**CellMLB**.***InterRatMlbUeNumThd***	Network plan (negotiation not required)	This parameter specifies the threshold for the number of UL-sync UEs in the cell used to evaluate triggering of inter-RAT MLB. A larger value of this parameter results in a lower probability of triggering MLB to UTRAN and a greater decrease in data transmission delay after MLB. However, a larger value leads to a longer period before MLB during which data transmission delay is long because of resource insufficiency. If this parameter is set too small, UEs are transferred even when the serving cell can provide high-quality services, leading to more signaling interaction and handovers. This increases the CPU load and the MLB algorithm produces little gains.

Copyright © Huawei Technologies Co., Ltd.

Parameter Name	Parameter ID	Data Source	Setting Notes
InterRAT MLB UE Select PRB Threshold	**CellMLB**.***InterRatMlbUeSelPrbThd***	Network plan (negotiation not required)	This parameter specifies the UE PRB usage threshold for UE selection in inter-RAT MLB. If the **CellMLB**.***InterRatMlbUeSelStrategy*** parameter is set to **PRBRATIOLOWMODE(PRB Ratio Low Mode)**, the eNodeB selects UEs whose PRB usage is less than the **CellMLB**.***InterRatMlbUeSelPrbThd*** parameter value.

(Optional) The following table describes the parameters that must be set in **SpidCfg** MOs to configure SPIDs. An **SpidCfg** MO consists of parameters related to the inter-RAT mobility priority of UL-sync UEs and the SPID-specific inter-RAT MLB switch.

Parameter Name	Parameter ID	Data Source	Setting Notes
Spid	**SpidCfg**.***Spid***	Network plan (negotiation not required)	This parameter specifies an SPID. The value range is **1** to **256**.
RAT frequency priority indication	**SpidCfg**.***RatFreqPriorityInd***	Network plan (negotiation not required)	This parameter specifies whether to configure a frequency priority group.
RAT frequency priority group ID	**SpidCfg**.***RatFreqPriorityGroupId***	Network plan (negotiation not required)	This parameter uniquely identifies a frequency priority group. The value range is **0** to **255**.

Copyright © Huawei Technologies Co., Ltd.

Parameter Name	Parameter ID	Data Source	Setting Notes
InterRat Mlb Switch	**SpidCfg.*InterRatMlbSwitch***	Network plan (negotiation not required)	This parameter specifies whether to allow inter-RAT MLB for UEs with the SPID. The values **TRUE(TRUE)** and **FALSE(FALSE)** indicate that inter-RAT MLB is allowed and prohibited, respectively. This parameter takes effect only for UEs that will access the network after the parameter is set.

8.1.6 Activation

Using the CME to Perform Batch Configuration for Newly Deployed eNodeBs

Enter the values of the parameters listed in **Table 8-1** in a summary data file, which also contains other data for the new eNodeBs to be deployed.

Then, import the summary data file into the Configuration Management Express (CME) for batch configuration. For detailed instructions, see "Creating eNodeBs in Batches" in the initial configuration guide for the eNodeB, which is available in the eNodeB product documentation.

The summary data file may be a scenario-specific file provided by the CME or a customized file, depending on the following conditions:

- The MOs in **Table 8-1** are contained in a scenario-specific summary data file. In this situation, set the parameters in the MOs, and then verify and save the file.
- Some MOs in **Table 8-1** are not contained in a scenario-specific summary data file. In this situation, customize a summary data file to include the MOs before you can set the parameters.

Table 8-1 Parameters for MLB to UTRAN

MO	Sheet in the Summary Data File	Parameter Group	Remarks
CellAlgoSwitch	CellAlgoSwitch	MlbAlgoSwitch	This MO must be customized in a list-type sheet of the template.

Copyright © Huawei Technologies Co., Ltd.

MO	Sheet in the Summary Data File	Parameter Group	Remarks
CellMLB	CellMLB	LocalCellId, LoadOffset, InterRatMlbThd, InitValidPeriod, InterRatMlbUeNumThd, InterRatMlbUeNumOffset, InterRatMlbUeSelPrbThd	This MO must be customized in a list-type sheet of the template.
ENodeBAlgoSwitch	ENodeBAlgoSwitch	HoModeSwitch	This MO must be customized in a list-type sheet of the template.
SpidCfg (Optional)	SpidCfg	Spid, InterFreqMlbSwitch, InterRatMlbSwitch	This MO must be customized in a list-type sheet of the template.

Using the CME to Perform Batch Activation for Existing eNodeBs

This feature can be batch activated using the CME. For detailed operations, see the following section in the CME product documentation or online help: **Managing the CME** > **CME Guidelines** > **Enhanced Feature Management** > **Feature Operation and Maintenance**.

Using the CME to Perform Batch Configuration for Existing eNodeBs

Batch reconfiguration using the CME is the recommended method to activate a feature on existing eNodeBs. This method reconfigures all data, except neighbor relationships, for multiple eNodeBs in a single procedure. The procedure is as follows:

Step 1 After creating a planned data area, choose CME > Advanced > Customize Summary Data File (U2000 client mode), or choose Advanced > Customize Summary Data File (CME client mode), to customize a summary data file for batch reconfiguration.

NOTE

For context-sensitive help on a current task in the client, press **F1**.

Step 2 Choose CME > LTE Application > Export Data > Export Base Station Bulk Configuration Data (U2000 client mode), or choose LTE Application > Export Data > Export Base Station Bulk Configuration Data (CME client mode), to export the eNodeB data stored on the CME into the customized summary data file.

Step 3 In the summary data file, set the parameters in the MOs listed in **Table 8-1** and close the file.

Step 4 Choose CME > LTE Application > Import Data > Import Base Station Bulk Configuration Data (U2000 client mode), or choose LTE Application > Import Data > Import Base Station

Copyright © Huawei Technologies Co., Ltd.

Bulk Configuration Data (CME client mode), to import the summary data file into the CME, and then start the data verification.

Step 5 After data verification is complete, choose CME > Planned Area > Export Incremental Scripts (U2000 client mode), or choose Area Management > Planned Area > Export Incremental Scripts (CME client mode), to export and activate the incremental scripts.

----End

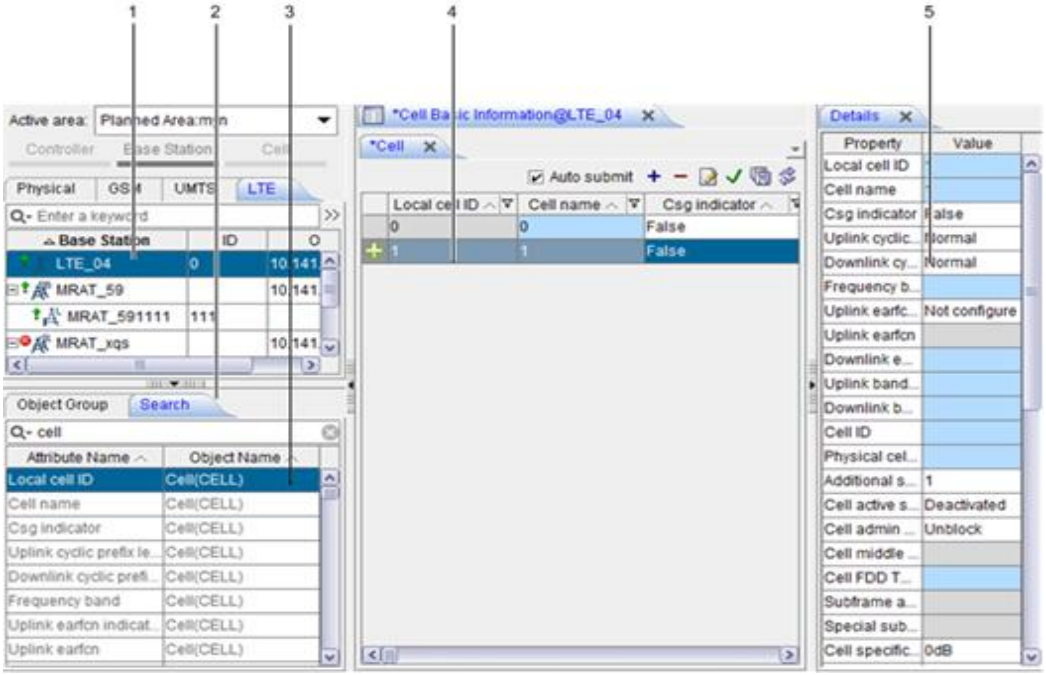
Using the CME to Perform Single Configuration

On the CME, set the parameters listed in the "Data Preparation" section for a single eNodeB. The procedure is as follows:

Step 1 In the planned data area, click **Base Station** in the upper left corner of the configuration window.

Step 2 In area 1 shown in **Figure 8-1**, select the eNodeB to which the MOs belong.

Figure 8-1 MO search and configuration window



Step 3 On the **Search** tab page in area 2, enter an MO name, for example, **CELL**.

Step 4 In area 3, double-click the MO in the **Object Name** column. All parameters in this MO are displayed in area 4.

Step 5 Set the parameters in area 4 or 5.

Step 6 Choose CME > Planned Area > Export Incremental Scripts (U2000 client mode), or choose Area Management > Planned Area > Export Incremental Scripts (CME client mode), to export and activate the incremental scripts.

----End

Copyright © Huawei Technologies Co., Ltd.

Using MML Commands

MLB to UTRAN by Transferring UL-Sync UEs

Step 1 Run the **MOD ENODEBALGOSWITCH** command with the **UtranPsHoSwitch(UtranPsHoSwitch)** option of the **Handover Mode switch** parameter selected.

Step 2 Run the **MOD CELLMLB** command to set the **Inter-RAT Mobility Load Balancing Threshold** and **Inter-RAT Mobility Load Balancing UE Number Threshold** parameters.

Step 3 Run the **MOD CELLALGOSWITCH** command with the **UtranMlbSwitch(UtranMlbSwitch)** option of the **Load balancing algorithm switch** parameter selected.

Step 4 (Optional) Run the **ADD SPIDCFG** command with the **InterRat Mlb Switch** parameter set to **TRUE(TRUE)** and other parameters specified for an SPID. If the SPID has already been configured, run the **MOD SPIDCFG** command to modify the configuration.

----End

MLB to UTRAN by Transferring RRC_REL UEs

Step 1 Run the **MOD CELLMLB** command to set the **Inter-RAT Mobility Load Balancing Threshold**, **Inter-RAT Idle MLB UE Number Threshold**, and **UTRAN Idle Mode Mobility Load Balancing Initial Valid Period** parameters.

Step 2 Run the **MOD CELLALGOSWITCH** command with the **UtranIdleMlbSwitch(UtranIdleMlbSwitch)** option of the **Load balancing algorithm switch** parameter selected.

----End

MML Command Examples

MLB to UTRAN by Transferring UL-Sync UEs

```
MOD ENODEBALGOSWITCH: HoModeSwitch=UtranPsHoSwitch-1;
MOD CELLMLB: LocalCellId=0, InterRatMlbThd=70, InterRatMlbUeNumThd=15;
MOD CELLALGOSWITCH: LocalCellId=0, MlbAlgoSwitch=UtranMlbSwitch-1;
ADD SPIDCFG: Spid=1, RatFreqPriorityInd=NOT_CFG, InterRatMlbSwitch=TRUE;
```

MLB to UTRAN by Transferring RRC_REL UEs

```
MOD CELLMLB: LocalCellId=0, InterRatMlbThd=70, InterRatIdleMlbUeNumThd=15,
InitValidPeriod=10;
MOD CELLALGOSWITCH: LocalCellId=0, MlbAlgoSwitch=UtranIdleMlbSwitch-1;
```

8.1.7 Activation Observation

MLB to UTRAN by Transferring UL-Sync UEs

To use SON logs to verify whether this MLB function has been activated, perform the following steps:

Step 1 On the U2000 client, choose **SON** > **SON Log**.

Step 2 On the **Query SON Log** tab page, click **LTE MLB Log** on the **Log Category** drop-down list in the upper left corner, and click **Inter-RAT Handover Statistics** in the **Custom** list under

Copyright © Huawei Technologies Co., Ltd.

Event Name. Then click **Query** to query SON logs. This MLB function has been activated if the following information is displayed in the event description:

- Load status of the serving cell: inter-RAT MLB threshold, uplink PRB usage, and downlink PRB usage
- Information about inter-RAT neighboring cells: MCC, MNC, RNC ID, UTRAN cell ID, and RAT
- Handover information: number of UEs that have been successfully handed over from the serving cell

----End

To use signaling tracing to verify whether this MLB function has been activated, perform the following steps:

Step 1 On the U2000 client, start a Uu interface tracing task, S1 interface tracing task, and a PRB usage monitoring task.

Step 2 Check the S1 interface tracing result for an S1AP_HANDOVER_REQUIRED message, as shown in **Figure 8-2**. If the S1AP_HANDOVER_REQUIRED message contains the cause value of "reduce-load-in-serving-cell", this MLB function has been activated.

Figure 8-2 HANDOVER REQUIRED message

----End

You can also observe performance counters to evaluate whether this MLB function has been activated. For the counters, see **8.1.10 Performance Monitoring**.

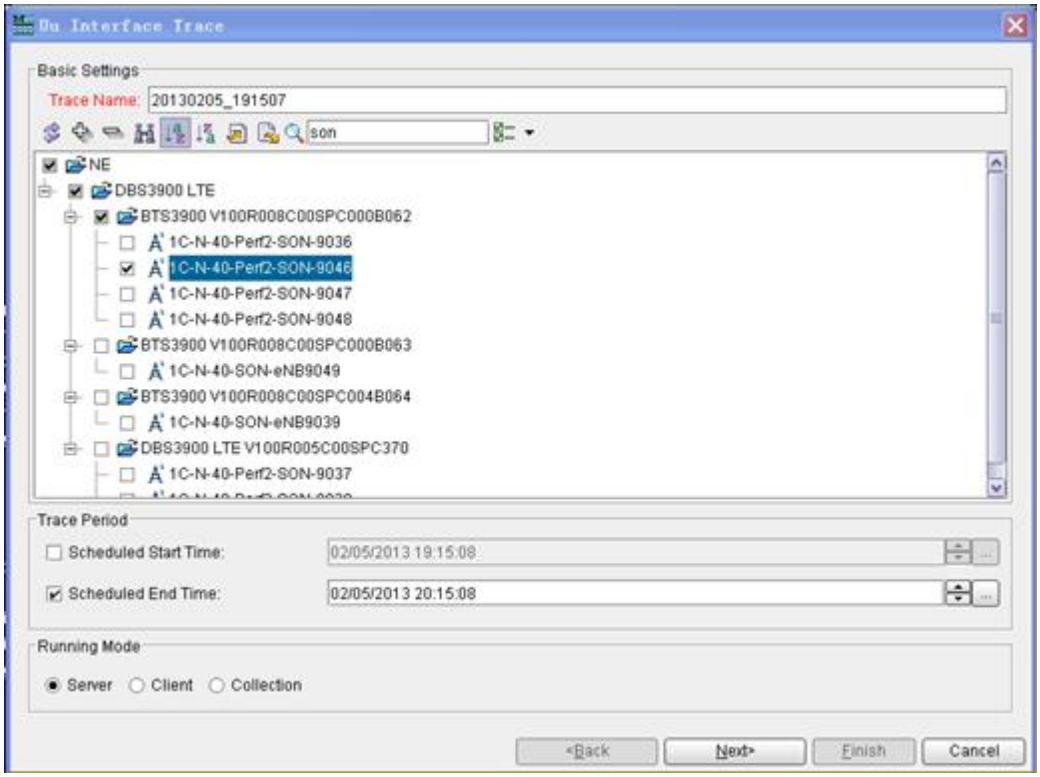
MLB to UTRAN by Transferring RRC_REL UEs

To verify whether this MLB function has been activated, start Uu interface tracing on the U2000 client as follows:

Copyright © Huawei Technologies Co., Ltd.

Step 1 In the navigation tree on the left of the **Signaling Trace Management** tab page, double-click **Uu Interface Trace** under **LTE** > **Application Layer**. The **Uu Interface Trace** dialog box is displayed, as shown in **Figure 8-3**.

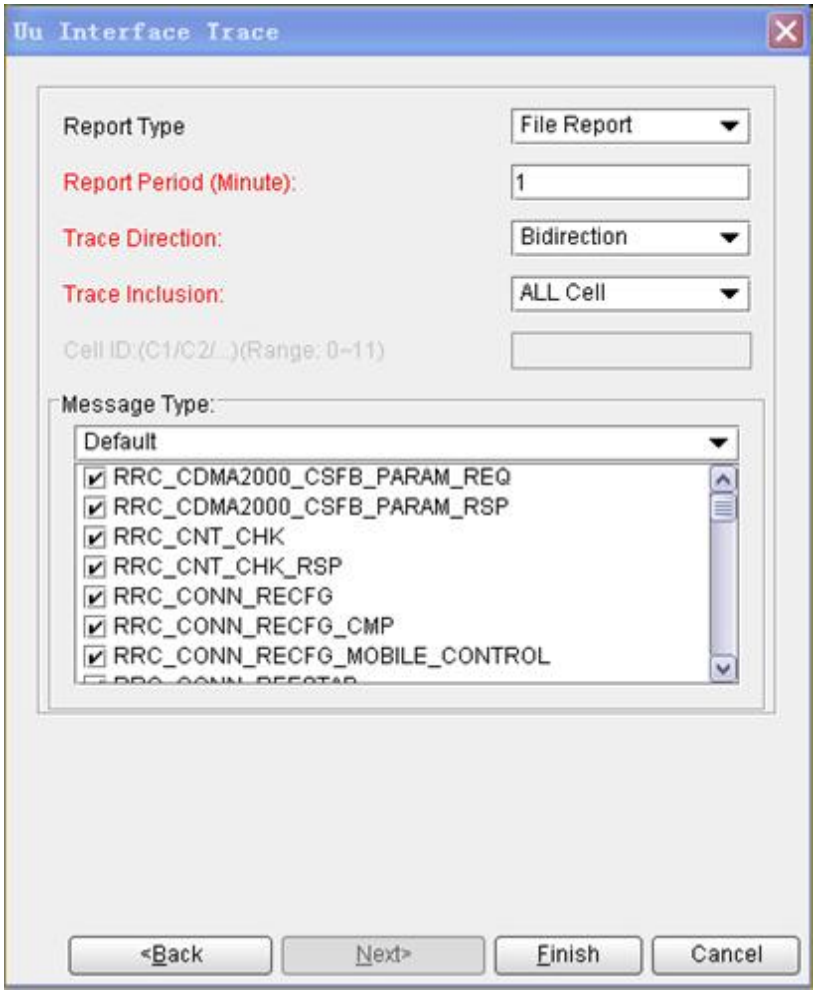
Figure 8-3 Uu Interface Trace dialog box



Step 2 Set the task name, select the eNodeB, and click **Next**. The **Uu Interface Trace** dialog box as shown in **Figure 8-4** is displayed. Then, click **Finish**.

Copyright © Huawei Technologies Co., Ltd.

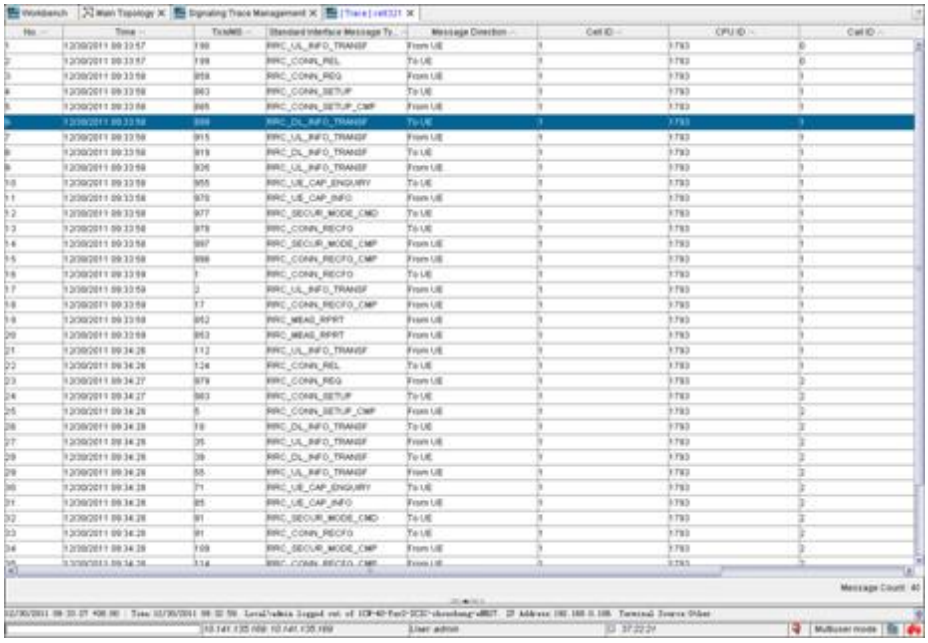
Figure 8-4 Entering information about a Uu interface tracing task



Step 3 After the Uu interface tracing task starts, view the tracing result, as shown in **Figure 8-5**.

Copyright © Huawei Technologies Co., Ltd.

Figure 8-5 Uu interface tracing result



This MLB function has been activated if the cell-reselection priority contained in the RRC_CONN_REL message in the Uu interface tracing result meets the requirements and the UE is successfully transferred to the UTRAN cell by cell reselection. For details about cell-reselection priorities, see **4.4.2 RRC_REL UEs**.

----End

8.1.8 Deactivation

Using the CME to Perform Batch Configuration

Batch reconfiguration using the CME is the recommended method to deactivate a feature on eNodeBs. This method reconfigures all data, except neighbor relationships, for multiple eNodeBs in a single procedure. The procedure for feature deactivation is similar to that for feature activation described in **Using the CME to Perform Batch Configuration for Existing eNodeBs**. In the procedure, modify parameters according to **Table 8-2**.

Copyright © Huawei Technologies Co., Ltd.

Table 8-2 Parameters for MLB to UTRAN

MO	Sheet in the Summary Data File	Parameter Group	Setting Notes
CellAlgoSwitch	CellAlgoSwitch	MlbAlgoSwitch	● To disable "MLB to UTRAN by transferring UL-sync UEs", deselect the **UtranMlbSwitch(UtranMlbSwitch)** option of this parameter. ● To disable "MLB to UTRAN by transferring RRC_REL UEs", deselect the **UtranIdleMlbSwitch(UtranIdleMlbSwitch)** option of this parameter.

Using the CME to Perform Single Configuration

On the CME, set parameters according to **Table 8-2**. For detailed instructions, see **Using the CME to Perform Single Configuration** described for feature activation.

Using MML Commands

MLB to UTRAN by Transferring UL-Sync UEs

Run the **MOD CELLALGOSWITCH** command with the **UtranMlbSwitch(UtranMlbSwitch)** option of the **Load balancing algorithm switch** parameter deselected.

MLB to UTRAN by Transferring RRC_REL UEs

Run the **MOD CELLALGOSWITCH** command with the **UtranIdleMlbSwitch(UtranIdleMlbSwitch)** option of the **Load balancing algorithm switch** parameter deselected.

MML Command Examples

MLB to UTRAN by Transferring UL-Sync UEs

```
MOD CELLALGOSWITCH: LocalCellId=0, MlbAlgoSwitch=UtranMlbSwitch-0;
```

MLB to UTRAN by Transferring RRC_REL UEs

```
MOD CELLALGOSWITCH: LocalCellId=0, MlbAlgoSwitch=UtranIdleMlbSwitch-0;
```


Huawei Proprietary and Confidential
Copyright © Huawei Technologies Co., Ltd.

8.1.9 Reconfiguration

For measurement-based handovers, set the **CellMLB**.***FreqSelectStrategy*** parameter to specify the policy that an eNodeB uses to select target frequencies. If the parameter is set to **PRIORITYBASED(PriorityBased)**, set the **EutranInterNFreq**.***MlbFreqPriority*** and **UtranNFreq**.***MlbFreqPriority*** parameters to specify the E-UTRAN and UTRAN frequency priorities based on which the eNodeB selects target frequencies. In addition, if both neighboring E-UTRAN frequencies and neighboring UTRAN frequencies are defined, it is required that the **CellMLB**.***InterFreqLoadEvalPrd*** and **CellMLB**.***InterRatLoadEvalPrd*** parameters be set to the same value.

8.1.10 Performance Monitoring

Table 8-3 lists the counters used to monitor the performance of MLB to UTRAN.

Table 8-3 Performance counters related to the feature of MLB to UTRAN

Counter ID	Counter Name	Counter Description
1526728529	L.IRATHO.E2W.Load.PrepAttOut	Number of load-based EUTRAN-to-WCDMA handover preparation attempts
1526728530	L.IRATHO.E2W.Load.ExecAttOut	Number of load-based EUTRAN-to-WCDMA handover execution attempts
1526728531	L.IRATHO.E2W.Load.ExecSuccOut	Number of successful load-based EUTRAN-to-WCDMA handover executions
1526728535	L.RRCRedirection.E2W.Load	Number of load-based EUTRAN-to-WCDMA redirections
1526728443	L.RRCRel.DedicatedPri.WCDMA.High	Number of times WCDMA frequencies are assigned the highest dedicated priority
1526729878	L.RRCRel.load.DedicatedPri.WCDMA.High	Number of times WCDMA frequencies are assigned the highest dedicated priority when the cell load is heavy

NOTE

Deployment of the features LOFD-00105401 Camp & Handover Based on SPID and LOFD-001112 MOCN Flexible Priority Based Camping has an impact on the value of the L.RRCRel.DedicatedPri.WCDMA.High counter.

Huawei Proprietary and Confidential
Copyright © Huawei Technologies Co., Ltd.

8.1.11 Parameter Optimization

The following parameters may need to be adjusted for better performance:

- **CellMLB**.***InterRatMlbThd***

  This parameter determines the probabilities and effect of inter-RAT MLB.
- **CellMLB**.***InterRatMlbUeNumOffset***

  A larger value of this parameter results in a lower probability of ping-pong load transfer.
- **CellMLB**.***InterRatMlbUeNumThd***

  This parameter controls the number of load-based E-UTRAN-to-UTRAN handovers and the amount of load that is transferred. A smaller value of this parameter results in a higher probability of triggering MLB to UTRAN. A larger value leads to a lower probability.
- **InterRatHoUtranGroup**.***LdSvBasedHoUtranB1ThdRscp*** or **InterRatHoUtranGroup**.***LdSvBasedHoUtranB1ThdEcn0***

  These two parameters control the signal level threshold and signal quality threshold for target UTRAN cells in handovers.
- **CellMLB**.***LoadDiffThd***

  This parameter controls the number and PRB usage of UEs that can be transferred within an MLB procedure. A larger value of this parameter results in a larger number of UEs that can be selected for load transfer within an MLB procedure.

8.1.12 Troubleshooting

None

8.2 LOFD-001045 Inter-RAT Load Sharing to GERAN

8.2.1 When to Use LOFD-001045

Use LOFD-001045 Inter-RAT Load Sharing to GERAN when neighboring GERAN cells but not neighboring UTRAN cells provide the same coverage as E-UTRAN cells.

8.2.2 Required Information

Collect the following information:

- Information about each neighboring cell of the cells served by the local eNodeB
  - Whether information about the neighboring cell is complete
  - Whether the neighboring cell has been blacklisted
  - Whether the **No handover indicator** parameter is set to **PERMIT_HO_ENUM(Permit Ho)**
- Status of the X2 interfaces with neighboring eNodeBs
- UE capabilities

  The proportion of UEs that support GERAN measurements
- Whether the operator has purchased the required licenses

Copyright © Huawei Technologies Co., Ltd.

8.2.3 Planning

8.2.3.1 RF Planning

For details, see **8.1.3.1 RF Planning**.

8.2.3.2 Network Planning

N/A

8.2.3.3 Hardware Planning

N/A

8.2.4 Requirements

There are no requirements for the operating system and transmission networking. Before deploying LOFD-001045 Inter-RAT Load Sharing to GERAN, the operator must purchase and activate the license for the feature listed in the following table.

Feature ID	Feature Name	License Control Item	NE	Sales Unit
LOFD-001045	Inter-RAT Load Sharing to GERAN	Inter-RAT Load Sharing to GERAN (FDD)	eNodeB	per RRC Connected User

8.2.5 Data Preparation

This section describes the data that you need to collect for setting parameters. Required data is data that you must collect for all scenarios. Collect scenario-specific data when necessary for a specific feature deployment scenario.

There are three types of data sources:

- Network plan (negotiation not required): parameter values planned and set by the operator
- Network plan (negotiation required): parameter values planned by the operator and negotiated with the EPC or peer transmission equipment
- User-defined: parameter values set by users

Required Data

The following table describes the parameter that must be set in a **CellMLB** MO to control the MLB algorithm for a cell.

Copyright © Huawei Technologies Co., Ltd.

Parameter Name	Parameter ID	Data Source	Setting Notes
Load Offset	**CellMLB**.***LoadOffset***	Network plan (negotiation not required)	This parameter specifies the offset applied to the threshold for triggering MLB. This mechanism helps prevent load fluctuations from frequently triggering and stopping MLB. The recommended value is **8**.

The following table describes the parameter that must be set in a **CellAlgoSwitch** MO to specify the handover policy for MLB from a cell.

Parameter Name	Parameter ID	Data Source	Setting Notes
Load Balancing Handover Policy	**CellAlgoswitch**.***MlbHoMode***	Network plan (negotiation not required)	If the **InterRatMlbBlind-Ho(InterRatMlbBlindHo)** option of this parameter is selected, the eNodeB preferentially uses blind inter-RAT handovers for load transfer to UTRAN or GERAN cells. If this option is not selected, blind inter-RAT handovers are disabled.

The following table describes the parameter that must be set in an **InterRatHoGeranGroup** MO to control inter-RAT handovers from an E-UTRAN cell to GERAN.

Copyright © Huawei Technologies Co., Ltd.

Parameter Name	Parameter ID	Data Source	Setting Notes
Load Service Based Geran EventB1 trigger threshold	**InterRatHoGeranGroup**.***LdSvBasedHoGeranB1Thd***	Network plan (negotiation not required)	This parameter specifies the RSSI threshold for event B1 related to load- or service-based inter-RAT handovers to GERAN. When the measured RSSI exceeds this threshold, a UE sends a B1 measurement report to the eNodeB. The recommended value is **-98**.

Scenario-specific Data

The following table describes the parameter that must be set in a **CellAlgoSwitch** MO to enable MLB to GERAN.

Parameter Name	Parameter ID	Data Source	Setting Notes
Load balancing algorithm switch	**CellAlgoSwitch**.***MlbAlgoSwitch***	Network plan (negotiation not required)	The **GeranMlbSwitch(GeranMlbSwitch)** option of this parameter specifies whether to enable MLB to GERAN. This algorithm is enabled only if this option is selected.

The following table describes the parameter that must be set in the **ENodeBAlgoSwitch** MO to specify handover policies for MLB to GERAN.

Copyright © Huawei Technologies Co., Ltd.

Parameter Name	Parameter ID	Data Source	Setting Notes
Handover Mode switch	**ENodeBAlgoSwitch**.***HoModeSwitch***	Network plan (negotiation not required)	If the **GeranPsHoSwitch(GeranPsHoSwitch)** option of this parameter is selected, the eNodeB can transfer UEs to neighboring GERAN cells through handovers. If this option is not selected, handovers to GERAN are disabled.

The following table describes the parameters that must be set in a **CellMLB** MO to configure the inter-RAT MLB algorithm for a cell.

Parameter Name	Parameter ID	Data Source	Setting Notes
Inter-RAT Mobility Load Balancing Threshold	**CellMLB**.***InterRatMlbThd***	Network plan (negotiation not required)	This parameter specifies the PRB usage threshold for inter-RAT MLB. The default value is **75**.
Inter-RAT Mobility Load Balancing UE Number Threshold	**CellMLB**.***InterRatMlbUeNumThd***	Network plan (negotiation not required)	This parameter specifies the threshold for the number of UL-sync UEs in the cell used to evaluate triggering of inter-RAT MLB. The actual threshold equals the percentage value of this parameter multiplied by 1000.

(Optional) The following table describes the parameters that must be set in **SpidCfg** MOs to configure SPIDs. An **SpidCfg** MO consists of parameters related to the inter-RAT mobility priority of UL-sync UEs and the SPID-specific inter-RAT MLB switch.

Copyright © Huawei Technologies Co., Ltd.

Parameter Name	Parameter ID	Data Source	Setting Notes
Spid	**SpidCfg**.***Spid***	Network plan (negotiation not required)	This parameter specifies an SPID. The value range is **1** to **256**.
RAT frequency priority indication	**SpidCfg**.***RatFreqPriorityInd***	Network plan (negotiation not required)	This parameter specifies whether to configure a frequency priority group.
RAT frequency priority group ID	**SpidCfg**.***RatFreqPriorityGroupId***	Network plan (negotiation not required)	This parameter uniquely identifies a frequency priority group. The value range is **0** to **255**.
InterRat Mlb Switch	**SpidCfg**.***InterRatMlbSwitch***	Network plan (negotiation not required)	This parameter specifies whether to allow inter-RAT MLB for UEs with the SPID. The values **TRUE(TRUE)** and **FALSE(FALSE)** indicate that inter-RAT MLB is allowed and prohibited, respectively. This parameter takes effect only for UEs that will access the network after the parameter is set.

8.2.6 Activation

Using the CME to Perform Batch Configuration for Newly Deployed eNodeBs

Enter the values of the parameters listed in **Table 8-4** in a summary data file, which also contains other data for the new eNodeBs to be deployed.

Then, import the summary data file into the Configuration Management Express (CME) for batch configuration. For detailed instructions, see "Creating eNodeBs in Batches" in the initial configuration guide for the eNodeB, which is available in the eNodeB product documentation.

The summary data file may be a scenario-specific file provided by the CME or a customized file, depending on the following conditions:

Copyright © Huawei Technologies Co., Ltd.

- The MOs in Table 8-4 are contained in a scenario-specific summary data file. In this situation, set the parameters in the MOs, and then verify and save the file.
- Some MOs in Table 8-4 are not contained in a scenario-specific summary data file. In this situation, customize a summary data file to include the MOs before you can set the parameters.

Table 8-4 Parameters for MLB to GERAN

MO	Sheet in the Summary Data File	Parameter Group	Remarks
CellAlgoSwitch	CELLALGOSWITCH	MlbAlgoSwitch	This MO must be customized in a list-type sheet of the template.
CellAlgoSwitch	CellAlgoSwitch	MlbHoMode	This MO must be customized in a list-type sheet of the template.
CellMLB	CELLMLB	LocalCellId, LoadOffset, InterRatMlbThd, InterRatMlbUeNumThd	This MO must be customized in a list-type sheet of the template.
SpidCfg (Optional)	SPIDCFG	Spid, InterFreqMlbSwitch, InterRatMlbSwitch	This MO must be customized in a list-type sheet of the template.

Using the CME to Perform Batch Activation for Existing eNodeBs

This feature can be batch activated using the CME. For detailed operations, see the following section in the CME product documentation or online help: **Managing the CME** > **CME Guidelines** > **Enhanced Feature Management** > **Feature Operation and Maintenance**.

Using the CME to Perform Batch Configuration for Existing eNodeBs

Batch reconfiguration using the CME is the recommended method to activate a feature on existing eNodeBs. This method reconfigures all data, except neighbor relationships, for multiple eNodeBs in a single procedure. The procedure is as follows:

Step 1 After creating a planned data area, choose CME > Advanced > Customize Summary Data File (U2000 client mode), or choose Advanced > Customize Summary Data File (CME client mode), to customize a summary data file for batch reconfiguration.

NOTE

For context-sensitive help on a current task in the client, press **F1**.

Step 2 Choose CME > LTE Application > Export Data > Export Base Station Bulk Configuration Data (U2000 client mode), or choose LTE Application > Export Data > Export Base Station

Copyright © Huawei Technologies Co., Ltd.

Bulk Configuration Data (CME client mode), to export the eNodeB data stored on the CME into the customized summary data file.

Step 3 In the summary data file, set the parameters in the MOs listed in **Table 8-4** and close the file.

Step 4 Choose CME > LTE Application > Import Data > Import Base Station Bulk Configuration Data (U2000 client mode), or choose LTE Application > Import Data > Import Base Station Bulk Configuration Data (CME client mode), to import the summary data file into the CME, and then start the data verification.

Step 5 After data verification is complete, choose CME > Planned Area > Export Incremental Scripts (U2000 client mode), or choose Area Management > Planned Area > Export Incremental Scripts (CME client mode), to export and activate the incremental scripts.

----End

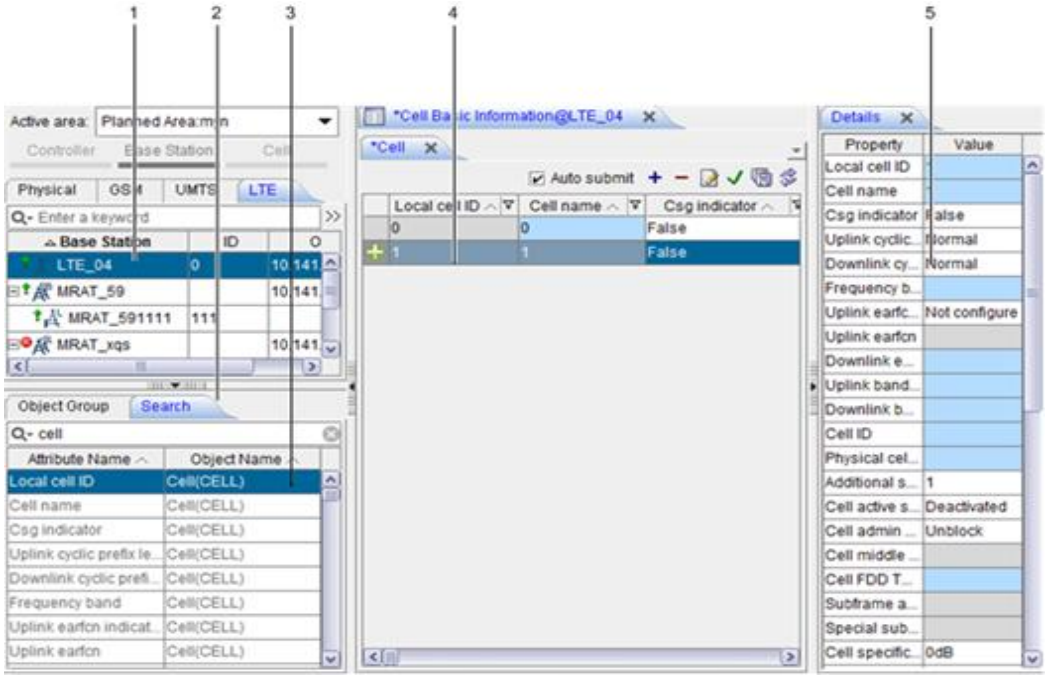
Using the CME to Perform Single Configuration

On the CME, set the parameters listed in the "Data Preparation" section for a single eNodeB. The procedure is as follows:

Step 1 In the planned data area, click **Base Station** in the upper left corner of the configuration window.

Step 2 In area 1 shown in **Figure 8-6**, select the eNodeB to which the MOs belong.

Figure 8-6 MO search and configuration window



Step 3 On the **Search** tab page in area 2, enter an MO name, for example, **CELL**.

Step 4 In area 3, double-click the MO in the **Object Name** column. All parameters in this MO are displayed in area 4.

Step 5 Set the parameters in area 4 or 5.

Copyright © Huawei Technologies Co., Ltd.

Step 6 Choose CME > Planned Area > Export Incremental Scripts (U2000 client mode), or choose Area Management > Planned Area > Export Incremental Scripts (CME client mode), to export and activate the incremental scripts.

----End

Using MML Commands

Step 1 Run the **MOD ENODEBALGOSWITCH** command with the **GeranPsHoSwitch(GeranPsHoSwitch)** option of the **Handover Mode switch** parameter selected.

Step 2 Run the **MOD CELLMLB** command to set the **Inter-RAT Mobility Load Balancing Threshold** and **Inter-RAT Mobility Load Balancing UE Number Threshold** parameters.

Step 3 Run the **MOD CELLALGOSWITCH** command with the **GeranMlbSwitch(GeranMlbSwitch)** option of the **Load balancing algorithm switch** parameter selected.

Step 4 (Optional) Run the **ADD SPIDCFG** command with the **InterRat Mlb Switch** parameter set to **TRUE(TRUE)** and other parameters specified for an SPID. If the SPID has already been configured, run the **MOD SPIDCFG** command to modify the configuration.

----End

MML Command Examples

```
MOD ENODEBALGOSWITCH: HoModeSwitch=GeranPsHoSwitch-1;
MOD CELLMLB: LocalCellId=0, InterRatMlbThd=70, InterRatMlbUeNumThd=15;
MOD CELLALGOSWITCH: LocalCellId=0, MlbAlgoSwitch=GeranMlbSwitch-1;
ADD SPIDCFG: Spid=1, RatFreqPriorityInd=NOT_CFG, InterRatMlbSwitch=TRUE;
```

8.2.7 Activation Observation

To use SON logs to verify whether MLB to GERAN has been activated, perform the following steps:

Step 1 On the U2000 client, choose **SON** > **SON Log**.

Step 2 On the **Query SON Log** tab page, click **LTE MLB Log** on the **Log Category** drop-down list in the upper left corner, and click **Inter-RAT Handover Statistics** in the **Custom** list under **Event Name**. Then click **Query** to query SON logs. MLB to GERAN has been activated if the following information is displayed in the event description:

- Load status of the serving cell: inter-RAT MLB threshold, uplink PRB usage, and downlink PRB usage
- Information about inter-RAT neighboring cells: MCC, MNC, location area code (LAC), GERAN cell ID, and RAT
- Handover information: number of UEs that have been successfully handed over from the serving cell

----End

You can also observe performance counters to evaluate whether MLB to GERAN has been activated. For the counters, see **8.2.9 Performance Monitoring**.

Copyright © Huawei Technologies Co., Ltd.

8.2.8 Deactivation

Using the CME to Perform Batch Configuration

Batch reconfiguration using the CME is the recommended method to deactivate a feature on eNodeBs. This method reconfigures all data, except neighbor relationships, for multiple eNodeBs in a single procedure. The procedure for feature deactivation is similar to that for feature activation described in **Using the CME to Perform Batch Configuration for Existing eNodeBs**. In the procedure, modify parameters according to **Table 8-5**.

Table 8-5 Parameters for MLB to GERAN

MO	Sheet in the Summary Data File	Parameter Group	Setting Notes
CellAlgoSwitch	CellAlgoSwitch	MlbAlgoSwitch	To disable the feature of MLB to GERAN, deselect the **GeranMlbSwitch(GeranMlbSwitch)** option of this parameter.

Using the CME to Perform Single Configuration

On the CME, set parameters according to **Table 8-5**. For detailed instructions, see **Using the CME to Perform Single Configuration** described for feature activation.

Using MML Commands

Run the **MOD CELLALGOSWITCH** command with the **GeranMlbSwitch(GeranMlbSwitch)** option of the **Load balancing algorithm switch** parameter deselected.

MML Command Examples

```
MOD CELLALGOSWITCH: LocalCellId=0, MlbAlgoSwitch=GeranMlbSwitch-0;
```

8.2.9 Performance Monitoring

Table 8-6 lists the counters used to monitor the performance of MLB to GERAN.

Table 8-6 Performance counters related to the feature of MLB to GERAN

Counter ID	Counter Name	Counter Description
1526728532	L.IRATHO.E2G.Load.PrepAttOut	Number of load-based EUTRAN-to-GERAN handover preparation attempts

Copyright © Huawei Technologies Co., Ltd.

Counter ID	Counter Name	Counter Description
1526728533	L.IRATHO.E2G.Load.ExecAttOut	Number of load-based EUTRAN-to-GERAN handover execution attempts
1526728534	L.IRATHO.E2G.Load.ExecSuccOut	Number of successful load-based EUTRAN-to-GERAN handover executions

8.2.10 Parameter Optimization

The following parameters may need to be adjusted for better performance:

- **CellMLB.*InterRatMlbThd***

  This parameter determines the probabilities and effect of inter-RAT MLB.
- **CellMLB.*InterRatMlbUeNumOffset***

  A larger value of this parameter results in a lower probability of ping-pong load transfer.

8.2.11 Troubleshooting

None

Copyright © Huawei Technologies Co., Ltd.

9 Parameters

Table 9-1 Parameters

MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
UtranNCell	OverlapInd	ADD UTRANNCELL MOD UTRANNCELL LST UTRANNCELL	LOFD-001044 / TDLOFD-001044 MRFD-231901	Inter-RAT Load Sharing to UTRAN Multi-RAT Carrier Joint Intelligent Shutdown(eNodeB)	Meaning: Indicates whether the neighboring UTRAN cell is manually identified as a neighboring cell overlapping the E-UTRAN cell. The default value is NO. The parameter value YES indicates that the neighboring UTRAN cell is manually identified as a neighboring cell overlapping the E-UTRAN cell. The parameter value NO indicates that the neighboring UTRAN cell is not manually identified as a neighboring cell overlapping the E-UTRAN cell. GUI Value Range: NO(NO), YES(YES) Unit: None Actual Value Range: NO, YES Default Value: NO(NO)
RrcConnStateTimer	UeInactiveTimer	MOD RRCCONNSTATETIMER LST RRCCONNSTATETIMER	LBFD-002007 / TDLBFD-002007	RRC Connection Management	Meaning: Indicates the length of the UE inactivity timer. If the eNodeB detects that a UE has neither received nor sent data for a duration exceeding the value of this parameter, the eNodeB releases the RRC connection for the UE. If this parameter is set to 0, the UE inactivity timer is not used. If this parameter is modified, the modified value applies only to UEs that access the network later. GUI Value Range: 0~3600 Unit: s Actual Value Range: 0~3600 Default Value: 20

Copyright © Huawei Technologies Co., Ltd.

MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
CellAlgoSwitch	MlbAlgoSwitch	MOD CELLALGOSWITCH LST CELLALGOSWITCH	LOFD-001032 / TDLOFD-001032 LOFD-001044 / TDLOFD-001044 LOFD-001045 / TDLOFD-001045 LOFD-001057	Intra-LTE Load Balancing Inter-RAT Load Sharing to UTRAN Inter-RAT Load Sharing to GERAN	Meaning: Indicates the switches used to enable or disable load balancing (MLB) algorithms. Switches specified by this parameter control the following algorithms: intra-frequency MLB, intra-frequency MLB for UEs in idle mode, inter-frequency MLB, inter-frequency MLB for UEs in idle mode, blind inter-frequency MLB, MLB to UTRAN, MLB to UTRAN for UEs in idle mode, MLB to GERAN, MLB to CDMA2000, MLB based on PRB evaluation, and MLB based on neighboring cell load status. If one switch is turned on, the corresponding algorithm is enabled to balance the loads between the serving cell and neighboring cells of the specified category. IntraFreqMlbSwitch: Indicates the switch used to enable or disable intra-frequency load balancing. If this parameter is turned on, intra-frequency load balancing is enabled and IntraFreqIdleMlbSwitch becomes valid. If this parameter is turned off, intra-frequency load balancing is disabled and IntraFreqIdleMlbSwitch does not take effect. InterFreqMlbSwitch: Indicates the switch used to enable or disable inter-frequency load balancing. UtranMlbSwitch: Indicates the switch used to enable or disable load sharing with UTRAN cells. GeranMlbSwitch: Indicates the switch used to enable or disable load sharing with GERAN cells. CdmaMlbSwitch: Indicates the switch used to enable or disable load sharing with CDMA2000 cells. IntraFreqIdleMlbSwitch: If both IntraFreqIdleMlbSwitch and IntraFreqMlbSwitch are turned on, intra-frequency load balancing for UEs in idle mode is enabled. Otherwise, intra-frequency idle load balancing for UEs in idle mode is disabled. UtranIdleMlbSwitch: Indicates the switch used to enable or disable load sharing with UTRAN for UEs in idle mode. MlbLoadInfoSwitch: Indicates the switch whether load of an inter-RAT cell is considered when the load sharing between the E-UTRAN cell and the inter-RAT cell is performed. In inter-RAT load sharing based on load of the inter-RAT cell, the inter-RAT cell must be a UTRAN cell. If both the MlbLoadInfoSwitch and UtranMlbSwitch options are selected, the UTRAN

Copyright © Huawei Technologies Co., Ltd.

MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
					cell whose load status is obtained and is normal or medium can be selected as the candidate cell for load sharing with UTRAN. If both the MlbLoadInfoSwitch and UtranIdleMlbSwitch options are selected, the UTRAN cell whose load status is obtained and is normal can be selected as the candidate cell for load sharing with UTRAN. InterFreqBlindMlbSwitch: Indicates the switch used to enable or disable the inter-frequency blind load balancing. PrbValMlbSwitch: Indicates the switch used to enable or disable the load balancing based on PRB evaluation. This switch takes effects only on LTE TDD cells. This switch will be removed in later versions. In this version, the setting of this switch is still synchronized between the U2000 and the eNodeB and can be delivered to the eNodeB, but it is no longer used internally. Therefore, you are advised to avoid using this switch. InterFreqIdleMlbSwitch: Indicates the switch used to enable or disable inter-frequency load balancing for UEs in idle mode. After the InterFreqIdleMlbSwitch option is selected, the load balancing for UEs in idle mode is valid only when at least one inter-frequency load balancing mode for UEs in idle mode is selected. GUI Value Range: IntraFreqMlbSwitch(IntraFreqMlbSwitch), InterFreqMlbSwitch(InterFreqMlbSwitch), UtranMlbSwitch(UtranMlbSwitch), GeranMlbSwitch(GeranMlbSwitch), CdmaMlbSwitch(CdmaMlbSwitch), IntraFreqIdleMlbSwitch(IntraFreqIdleMlbSwitch), UtranIdleMlbSwitch(UtranIdleMlbSwitch), MlbLoadInfoSwitch(MlbLoadInfoSwitch), InterFreqBlindMlbSwitch(InterFreqBlindMlbSwitch), PrbValMlbSwitch(PrbValMlbSwitch), InterFreqIdleMlbSwitch(InterFreqIdleMlbSwitch) Unit: None Actual Value Range: IntraFreqMlbSwitch, InterFreqMlbSwitch, UtranMlbSwitch, GeranMlbSwitch, CdmaMlbSwitch, IntraFreqIdleMlbSwitch, UtranIdleMlbSwitch, MlbLoadInfoSwitch, InterFreqBlindMlbSwitch, PrbValMlbSwitch, InterFreqIdleMlbSwitch Default Value: IntraFreqMlbSwitch:Off, InterFreqMlbSwitch:Off, UtranMlbSwitch:Off,

Copyright © Huawei Technologies Co., Ltd.

MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
					GeranMlbSwitch:Off, CdmaMlbSwitch:Off, IntraFreqIdleMlbSwitch:Off, UtranIdleMlbSwitch:Off, MlbLoadInfoSwitch:Off, InterFreqBlindMlbSwitch:Off, PrbValMlbSwitch:Off, InterFreqIdleMlbSwitch:Off

Copyright © Huawei Technologies Co., Ltd.

MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
ENodeBAlgoSwitch	HoModeSwitch	MOD ENODEBALGOSWITCH LST ENODEBALGOSWITCH	LOFD-001022 / TDLOFD-001022 LOFD-001023 / TDLOFD-001023 LOFD-001033 / TDLOFD-001033 LOFD-001034 / TDLOFD-001034 LOFD-001019 / TDLOFD-001019 LOFD-001020 / TDLOFD-001020 LOFD-001021 / TDLOFD-001021 TDLOFD-001052 TDLOFD-001088 TDLOFD-001043	SRVCC to UTRAN SRVCC to GERAN CS Fallback to UTRAN CS Fallback to GERAN PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and GERAN PS Inter-RAT Mobility between E-UTRAN and CDMA2000 Flash CS Fallback to UTRAN CS Fallback	Meaning: Indicates the switches corresponding to the inputs based on which the eNodeB determines handover policies. EutranVoipCapSwitch: UtranVoipCapSwitch: If this switch is turned on, UTRAN supports VoIP. If this switch is turned off, UTRAN does not support VoIP. GeranVoipCapSwitch: This switch will be removed in later versions. In this version, the setting of this switch is still synchronized between the U2000 and the eNodeB, but it is no longer used internally. Therefore, you are advised to avoid using this switch. Cdma1xRttVoipCapSwitch: If this switch is turned on, CDMA2000 1xRTT supports VoIP. If this switch is turned off, CDMA2000 1xRTT does not support VoIP. UtranPsHoSwitch: If this switch is turned on, UTRAN supports PS handovers. If this switch is turned off, UTRAN does not support PS handovers. GeranPsHoSwitch: If this switch is turned on, GERAN supports PS handovers. If this switch is turned off, GERAN does not support PS handovers. CdmaHrpdNonOtpimisedHoSwitch: If this switch is turned on, non-optimized handovers to CDMA2000 HRPD are enabled. If this switch is turned off, non-optimized handovers to CDMA2000 HRPD are disabled. CdmaHrpdOptimisedHoSwitch: If this switch is turned on, optimized handovers to CDMA2000 HRPD are enabled. If this switch is turned off, optimized handovers to CDMA2000 HRPD are disabled. GeranNaccSwitch: This switch does not take effect if GeranCcoSwitch is disabled. If this switch is turned on, the GERAN supports network assisted cell change (NACC). If this switch is turned off, the GERAN does not support NACC. GeranCcoSwitch: If this switch is turned on, the GERAN supports cell change order (CCO). If this switch is turned off, the GERAN does not support CCO. UtranSrvccSwitch: If this switch is turned on, the UTRAN supports single radio voice call continuity (SRVCC). If this switch is turned off, the UTRAN does not support SRVCC. GeranSrvccSwitch: If this switch is turned on, the GERAN supports SRVCC. If this switch is turned off, the GERAN does not support SRVCC. Cdma1xRttSrvccSwitch: If this switch is turned on, the CDMA2000 1xRTT supports SRVCC. If this switch is turned off, the CDMA2000 1xRTT does not support SRVCC. UtranRedirectSwitch: If this switch is turned on, redirection to UTRAN is enabled. If this switch is turned off, redirection to UTRAN is disabled. GeranRedirectSwitch: If this switch is turned

Copyright © Huawei Technologies Co., Ltd.

MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
			TDLOFD-001072 TDLOFD-001046 TDLOFD-001073	Steering to UTRAN Service based Inter-RAT handover to UTRAN Distance based Inter-RAT handover to UTRAN Service based Inter-RAT handover to GERAN Distance based Inter-RAT handover to GERAN	on, redirection to GERAN is enabled. If this switch is turned off, redirection to GERAN is disabled. CdmaHrpdRedirectSwitch: If this switch is turned on, redirection to CDMA2000 HRPD is enabled. If this switch is turned off, redirection to CDMA2000 HRPD is disabled. Cdma1xRttRedirectSwitch: If this switch is turned on, redirection to CDMA2000 1xRTT is enabled. If this switch is turned off, redirection to CDMA2000 1xRTT is disabled. BlindHoSwitch: If this switch is turned on, blind handovers for CSFB are enabled. If this switch is turned off, blind handovers for CSFB are disabled. LcsSrvccSwitch: If this switch is turned on, an SRVCC procedure is triggered when a UE receives a CSFB instruction during a VoIP service. If this switch is turned off, an SRVCC procedure is not triggered when a UE receives a CSFB instruction during a VoIP service. AutoGapSwitch: If this switch is turned on and UEs support automatic measurement gap configurations on the target frequency, the eNodeB does not deliver gap configurations to UEs. If this switch is turned off, the eNodeB delivers gap configurations to UEs during all inter-frequency and inter-RAT measurements. UeVoipOnHspaCapSwitch: If this switch is turned on and the eNodeB attempts to hand over UEs using voice services to UTRAN, the eNodeB checks UE capabilities when determining whether PS handover is applied. UEs must support voiceOverPS-HS-UTRA-FDD-r9 if the target UTRAN cell works in FDD mode or voiceOverPS-HS-UTRA-TDD128-r9 if the target UTRAN cell works in TDD mode. If this switch is turned off, the eNodeB does not check UE capabilities when handing over UEs to UTRAN based on PS handovers. UtranFddB1CapSwitch: If this switch is turned on, the setting of bit 41 of FGI specifying the UE capability of event B1 measurement on FDD UTRAN cells must be considered. If this switch is turned off, the setting of bit 41 of FGI does not need to be considered. GUI Value Range: EutranVoipCapSwitch(EutranVoipCapSwitch), UtranVoipCapSwitch(UtranVoipCapSwitch), GeranVoipCapSwitch(GeranVoipCapSwitch), Cdma1xRttVoipCapSwitch(Cdma1xRttVoipCapSwitch), UtranPsHoSwitch(UtranPsHoSwitch), GeranPsHoSwitch(GeranPsHoSwitch), CdmaHrpdNonOtpimisedHoSwitch(CdmaHrpdNonOtpimisedHoSwitch), CdmaHrpdOptimisedHoSwitch(CdmaHrpdOptimised-

Copyright © Huawei Technologies Co., Ltd.

MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
					HoSwitch), GeranNaccSwitch(GeranNaccSwitch), GeranCcoSwitch(GeranCcoSwitch), UtranSrvccSwitch(UtranSrvccSwitch), GeranSrvccSwitch(GeranSrvccSwitch), Cdma1xRttSrvccSwitch(Cdma1xRttSrvccSwitch), UtranRedirectSwitch(UtranRedirectSwitch), GeranRedirectSwitch(GeranRedirectSwitch), CdmaHrpdRedirectSwitch(CdmaHrpdRedirectS-witch), Cdma1xRttRedirectSwitch(Cdma1xRttRedirectSwitch), BlindHoSwitch(BlindHoSwitch), LcsSrvccSwitch(LcsSrvccSwitch), AutoGapSwitch(AutoGapSwitch), UeVoipOnHspaCapSwitch(UeVoipOnHspaCapS-witch), UtranFddB1CapSwitch(UtranFddB1CapSwitch) Unit: None Actual Value Range: EutranVoipCapSwitch, UtranVoipCapSwitch, GeranVoipCapSwitch, Cdma1xRttVoipCapSwitch, UtranPsHoSwitch, GeranPsHoSwitch, CdmaHrpdNonOtpimisedHoS-witch, CdmaHrpdOptimisedHoSwitch, GeranNaccSwitch, GeranCcoSwitch, UtranSrvccSwitch, GeranSrvccSwitch, Cdma1xRttSrvccSwitch, UtranRedirectSwitch, GeranRedirectSwitch, CdmaHrpdRedirectSwitch, Cdma1xRttRedirectSwitch, BlindHoSwitch, LcsSrvccSwitch, AutoGapSwitch, UeVoipOnHspaCapSwitch, UtranFddB1CapSwitch Default Value: EutranVoipCapSwitch:On, UtranVoipCapSwitch:Off, GeranVoipCapSwitch:Off, Cdma1xRttVoipCapSwitch:Off, UtranPsHoSwitch:Off, GeranPsHoSwitch:Off, CdmaHrpdNonOtpimisedHoSwitch:Off, CdmaHrpdOptimisedHoSwitch:Off, GeranNaccSwitch:Off, GeranCcoSwitch:Off, UtranSrvccSwitch:Off, GeranSrvccSwitch:Off, Cdma1xRttSrvccSwitch:Off, UtranRedirectSwitch:Off, GeranRedirectSwitch:Off, CdmaHrpdRedirectSwitch:Off, Cdma1xRttRedirectSwitch:Off, BlindHoSwitch:Off, LcsSrvccSwitch:Off, AutoGapSwitch:Off, UeVoipOnHspaCapSwitch:Off, UtranFddB1CapSwitch:Off

Copyright © Huawei Technologies Co., Ltd.

MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
CellMLB	InterRatMlbThd	MOD CELLMLB LST CELLMLB	LOFD-001044 / TDLOFD-001044 LOFD-001045 / TDLOFD-001045	Inter-RAT Load Sharing to UTRAN Inter-RAT Load Sharing to GERAN	Meaning: Indicates the threshold of the PRB usage for triggering inter-RAT load sharing. Inter-RAT load sharing is triggered if the number of synchronized UEs in the cell is greater than or equal to the sum of the InterRatMlbUeNumThd and InterRatMlbUeNumOffset parameter values and the PRB usage of the cell is greater than or equal to the InterRatMlbThd and LoadOffset parameter values. Inter-RAT load sharing stops if the number of synchronized UEs in the cell is less than the InterRatMlbUeNumThd parameter value or the PRB usage of the cell is less than the InterRatMlbThd parameter value. GUI Value Range: 1~100 Unit: % Actual Value Range: 1~100 Default Value: 75

Copyright © Huawei Technologies Co., Ltd.

MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
CellMLB	LoadOffset	MOD CELLMLB LST CELLMLB	LOFD-001032 / TDLOFD-001032 LOFD-001044 / TDLOFD-001044 LOFD-001045 / TDLOFD-001045	Intra-LTE Load Balancing Inter-RAT Load Sharing to UTRAN Inter-RAT Load Sharing to GERAN	Meaning: Indicates the offset of the PRB usage for triggering inter-frequency load balancing. Inter-frequency load balancing is triggered if the number of synchronized UEs in the cell is greater than or equal to the sum of the MlbMinUeNumThd and MlbMinUeNumOffset parameter values and the PRB usage of the cell is greater than or equal to the sum the InterFreqMlbThd and LoadOffset parameter values. Inter-frequency load balancing is triggered if the number of synchronized UEs in the cell is greater than or equal to the sum of the MlbMinUeNumThd and MlbMinUeNumOffset parameter values and the PRB valuation in the cell is greater than or equal to the sum of the PrbValMlbTrigThd and LoadOffset parameter values. Inter-RAT load sharing is triggered if the number of synchronized UEs in the cell is greater than or equal to the sum of the InterRatMlbUeNumThd and InterRatMlbUeNumOffset parameter values and the PRB usage of the cell is greater than or equal to the InterRatMlbThd and LoadOffset parameter values. Inter-RAT load sharing to transfer to-be-released UEs is triggered if the number of synchronized UEs is greater than or equal to the sum of the InterRATIdleMlbUeNumThd and InterRatMlbUeNumOffset parameter values and the PRB usage of the cell is greater than or equal to the sum of the InterRatMlbThd and LoadOffset parameter values. GUI Value Range: 0~50 Unit: % Actual Value Range: 0~50 Default Value: 8

Copyright © Huawei Technologies Co., Ltd.

MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
CellMLB	InterRatMlbUeNumThd	MOD CELLMLB LST CELLMLB	LOFD-001044 / TDLOFD-001044 LOFD-001045 / TDLOFD-001045	Inter-RAT Load Sharing to UTRAN Inter-RAT Load Sharing to GERAN	Meaning: Indicates the threshold of the number of synchronized UEs for triggering inter-RAT load sharing. Inter-RAT load sharing is triggered when the PRB usage of the cell is greater than or equal to the sum of the InterRatMlbThd and LoadOffset parameter values and the number of synchronized UEs in the cell is greater than or equal to the sum of the InterRatMlbUeNumThd and InterRatMlbUeNumOffset parameter values. Inter-RAT load sharing stops if the number of synchronized UEs in the cell is less than the InterRatMlbUeNumThd parameter value or the PRB usage of the cell is less than the InterRatMlbThd parameter value. The actual threshold of the number of synchronized UEs equals the percentage value multiplied by 1000.For example, parameter value 1 indicates that the threshold for the number of synchronized UEs is 10 (1000 x 1%), parameter value 2 indicates threshold 20, and parameter value 10 indicates threshold 100. Note that if the parameter value is 100 or 99, the threshold for the number of synchronized UEs is 1 or 2, respectively. GUI Value Range: 1~100 Unit: % Actual Value Range: 1~100 Default Value: 15

Copyright © Huawei Technologies Co., Ltd.

MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
CellMLB	InterRatMlbUeNumOffset	MOD CELLMLB LST CELLMLB	LOFD-001044 / TDLOFD-001044 LOFD-001045 / TDLOFD-001045 LOFD-070215 / TDLOFD-070215	Inter-RAT Load Sharing to UTRAN Inter-RAT Load Sharing to GERAN Intra-LTE User Number Load Balancing	Meaning: Indicates the offset of the number of synchronized UEs for triggering inter-RAT load sharing. Inter-RAT load sharing is triggered when the PRB usage of the cell is greater than or equal to the sum of the InterRatMlbThd and LoadOffset parameter values and the number of synchronized UEs in the cell is greater than or equal to the sum of the InterRatMlbUeNumThd and InterRatMlbUeNumOffset parameter values. Inter-RAT load sharing to transfer to-be-released UEs is triggered if the number of synchronized UEs is greater than or equal to the sum of the InterRATIdleMlbUeNumThd and InterRatMlbUeNumOffset parameter values and the PRB usage of the cell is greater than or equal to the sum of the InterRatMlbThd and LoadOffset parameter values. GUI Value Range: 0~10000 Unit: None Actual Value Range: 0~10000 Default Value: 0
CellMLB	InterRatIdleMlbUeNumThd	MOD CELLMLB LST CELLMLB	LOFD-001044/ TDLOFD-001044 LOFD-001045/ TDLOFD-001045	Inter-RAT Load Sharing to UTRAN Inter-RAT Load Sharing to GERAN	Meaning: Indicates the threshold of the number of synchronized UEs for triggering inter-RAT load sharing to transfer to-be-released UEs. Inter-RAT load sharing to transfer to-be-released UEs is triggered if the number of synchronized UEs is greater than or equal to the sum of the InterRATIdleMlbUeNumThd and InterRatMlbUeNumOffset parameter values and the PRB usage of the cell is greater than or equal to the sum of the InterRatMlbThd and LoadOffset parameter values. Inter-RAT load sharing to transfer to-be-released UEs stops if the number of synchronized UEs is less than the InterRATIdleMlbUeNumThd parameter value or the PRB usage of the cell is less than of the InterRatMlbThd parameter value. GUI Value Range: 1~10000 Unit: None Actual Value Range: 1~10000 Default Value: 150

Copyright © Huawei Technologies Co., Ltd.

MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
CellMLB	InterRatLoadEvalPrd	MOD CELLMLB LST CELLMLB	LOFD-001044 / TDLOFD-001044 LOFD-001045 / TDLOFD-001045	Inter-RAT Load Sharing to UTRAN Inter-RAT Load Sharing to GERAN	Meaning: Indicates the evaluation period for determining whether to trigger inter-RAT load sharing. If the serving cell remains in the heavy load state within an evaluation period, inter-RAT load sharing is triggered and certain UEs are transferred to an inter-RAT neighboring cell. If the UtranMlbSwitch or GeranMlbSwitch option of the MlbAlgoSwitch parameter is selected, the InterFreqMlbSwitch or InterFreqBlindMlbSwitch option of the MlbAlgoSwitch parameter is selected, and the FreqSelectStrategy parameter is set to PRIORITYBASED(PriorityBased), the values of the InterRatLoadEvalPrd and InterFreqLoadEvalPrd parameters must be the same. If the two parameters are set to different values, the evaluation period specified by the InterRatLoadEvalPrd parameter is adapted when both cell-level inter-frequency load balancing and inter-RAT load sharing are triggered at the same time. GUI Value Range: 5~30 Unit: s Actual Value Range: 5~30 Default Value: 30
GlobalProcSwitch	UtranLoadTransChan	MOD GLOBALPROCSWITCH LST GLOBALPROCSWITCH	None	None	Meaning: Indicates the UMTS load transmission channel. The eNodeB sends RAN-INFORMATION-REQUEST PDUs to UTRAN cells to request multiple reports on the load status of UTRAN cells only when the parameter is set to BASED_ON_RIM.The function specified by the parameter value BASED_ON_ECO is temporarily unavailable. GUI Value Range: NULL, BASED_ON_RIM, BASED_ON_ECO Unit: None Actual Value Range: NULL, BASED_ON_RIM, BASED_ON_ECO Default Value: NULL

Copyright © Huawei Technologies Co., Ltd.

MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
ENodeBAlgoSwitch	RimOnEcoSwitch	MOD ENODEBALGOSWITCH LST ENODEBALGOSWITCH	MRFD-090211 LOFD-001052/TDLOFD-001052 LOFD-001019/TDLOFD-001019 LOFD-001044/TDLOFD-001044 LOFD-001033/TDLOFD-001033	eCoordinator based RIM process optimization Flash CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN Inter-RAT Load Sharing to UTRAN(based on UMTS cell load information) CS Fallback to UTRAN (based on UMTS cell load information)	Meaning: Indicates whether the RAN information management (RIM) procedure is initiated by the eCoordinator. If this parameter is set to ON, the RIM procedure is initiated by the eCoordinator. If this parameter is set to OFF, the RIM procedure is initiated by the core network. GUI Value Range: OFF(Off), ON(On) Unit: None Actual Value Range: OFF, ON Default Value: OFF(Off)

Copyright © Huawei Technologies Co., Ltd.

MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
UtranNCell	NoHoFlag	ADD UTRANNCELL MOD UTRANNCELL LST UTRANNCELL	LOFD-001019 / TDLOFD-001019	PS Inter-RAT Mobility between E-UTRAN and UTRAN	Meaning: Indicates whether to allow handover of UEs to the neighboring cell that is determined by the neighboring relation. GUI Value Range: PERMIT_HO_ENUM(Permit Ho), FORBID_HO_ENUM(Forbid Ho) Unit: None Actual Value Range: PERMIT_HO_ENUM, FORBID_HO_ENUM Default Value: PERMIT_HO_ENUM(Permit Ho)
CellMLB	FreqSelectStrategy	MOD CELLMLB LST CELLMLB	LOFD-070215 / TDLOFD-070215	Intra-LTE User Number Load Balancing	Meaning: Indicates the target frequency selection policy applied for inter-frequency load balancing and inter-RAT load sharing when candidate cells work at different frequencies. If this parameter is set to FAIRSTRATEGY(FairStrategy), an E-UTRAN frequency among all neighboring E-UTRAN frequencies of candidate cells is randomly selected as the target frequency in inter-frequency load balancing, and all inter-RAT neighboring frequencies of candidate cells are selected as the target frequencies in inter-RAT load sharing. If this parameter is set to PRIORITYBASED(PriorityBased), the frequency with the highest priority among all intra-RAT neighboring frequencies and the frequency with the highest priority among all inter-RAT frequencies are selected as target frequencies separately when inter-frequency load balancing and inter-RAT load sharing are triggered separately or both operator-level inter-frequency load balancing and inter-RAT load sharing are triggered together; the frequency with the highest priority among all neighboring frequencies when both cell-level inter-frequency load balancing and inter-RAT load sharing are triggered together. GUI Value Range: FAIRSTRATEGY(FairStrategy), PRIORITYBASED(PriorityBased) Unit: None Actual Value Range: FAIRSTRATEGY, PRIORITYBASED Default Value: FAIRSTRATEGY(FairStrategy)

Copyright © Huawei Technologies Co., Ltd.

MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
UtranNFreq	MlbFreqPriority	ADD UTRANNFREQ MOD UTRANNFREQ LST UTRANNFREQ	LOFD-001032/ TDLOFD-001032	Intra-LTE Load Balancing	Meaning: Indicates the priority of the neighboring UTRAN frequency for load sharing with UTRAN. The value 0 indicates the lowest priority. The neighboring UTRAN frequency with the highest priority is selected as the target frequency for inter-RAT load sharing. GUI Value Range: 0~7 Unit: None Actual Value Range: 0~7 Default Value: 0
EutranInterNFreq	MlbFreqPriority	ADD EUTRANINTERNFREQ MOD EUTRANINTERNFREQ LST EUTRANINTERNFREQ	LOFD-001032/ TDLOFD-001032	Intra-LTE Load Balancing	Meaning: Indicates the priority of the neighboring E-UTRAN frequency for load balancing. The value 0 indicates the lowest priority. The neighboring E-UTRAN frequency with the highest priority is selected as the target frequency for inter-frequency load balancing. GUI Value Range: 0~7 Unit: None Actual Value Range: 0~7 Default Value: 7
SpidCfg	InterRatMlbSwitch	ADD SPIDCFG MOD SPIDCFG LST SPIDCFG	LOFD-00105401 / TDLOFD-00105401 LOFD-001044 / TDLOFD-001044 LOFD-001045 / TDLOFD-001045	Camp & Handover Based on SPID Inter-RAT Load Sharing to UTRAN Inter-RAT Load Sharing to GERAN	Meaning: Indicates whether to enable or disable inter-RAT load sharing for the SPID. The values TRUE or FALSE indicate that inter-RAT load sharing is allowed or prohibited for UEs with the SPID, respectively. GUI Value Range: FALSE(FALSE), TRUE(TRUE) Unit: None Actual Value Range: FALSE, TRUE Default Value: TRUE(TRUE)

Copyright © Huawei Technologies Co., Ltd.

MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
CellMLB	InterRatMlbUeSelStrategy	MOD CELLMLB LST CELLMLB	LOFD-001044 / TDLOFD-001044 LOFD-001045 / TDLOFD-001045	Inter-RAT Load Sharing to UTRAN Inter-RAT Load Sharing to GERAN	Meaning: Indicates the strategy for selecting UEs in inter-RAT load sharing triggered by the PRB usage. If this parameter is set to PRBRATIOHIGHMODE, the eNodeB transfers UEs whose PRB usage is greater than 2% but less than the half of the LoadDiffThd parameter value. If this parameter is set to PRBRATIOLOWMODE, the eNodeB transfers UEs whose PRB usage is less than or equal to the InterRatMlbUeSelPrbThd parameter value. GUI Value Range: PRBRATIOHIGHMODE(PRBRATIOHIGHMODE), PRBRATIOLOWMODE(PRBRATIOLOWMODE) Unit: None Actual Value Range: PRBRATIOHIGHMODE, PRBRATIOLOWMODE Default Value: PRBRATIOHIGHMODE(PRBRATIOHIGHMODE)
CellMLB	LoadDiffThd	MOD CELLMLB LST CELLMLB	LOFD-001032 / TDLOFD-001032 LOFD-001044 / TDLOFD-001044 LOFD-001045 / TDLOFD-001045	Intra-LTE Load Balancing Inter-RAT Load Sharing to UTRAN Inter-RAT Load Sharing to GERAN	Meaning: Indicates the threshold of PRB usage difference between the serving cell and a neighboring cell. When the result of the PRB usage of the serving cell minus that of a neighboring cell is greater than the parameter value, this neighboring cell can be a target cell for mobility load balancing (MLB). GUI Value Range: 1~50 Unit: % Actual Value Range: 1~50 Default Value: 15

Copyright © Huawei Technologies Co., Ltd.

MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
CellMLB	InterRatMlbUeSelPrbThd	MOD CELLMLB LST CELLMLB	LOFD-001044 / TDLOFD-001044 LOFD-001045 / TDLOFD-001045	Inter-RAT Load Sharing to UTRAN Inter-RAT Load Sharing to GERAN	Meaning: Indicates the PRB usage threshold for selecting UEs in inter-RAT load sharing triggered by the PRB usage. UEs whose PRB usage is less than or equal to this threshold are selected when the InterRatMlbUeSelStrategy parameter value is PRBRATIOLOWMODE. GUI Value Range: 0~50 Unit: % Actual Value Range: 0~50 Default Value: 2
InterRatHoUtranGroup	LdSvBasedHoUtranB1ThdRscp	ADD INTERRATHOUTRANGROUP MOD INTERRATHOUTRANGROUP LST INTERRATHOUTRANGROUP	LOFD-001043 TDLOFD-001022	Service based inter-RAT handover to UTRAN SRVCC to UTRAN	Meaning: Indicates the RSCP threshold for event B1 related to load- or service-based inter-RAT handover to UTRAN. This parameter specifies the requirement for RSCP of the target UTRAN cell. When the measurement value exceeds this threshold, a measurement report will be sent. GUI Value Range: -120~-25 Unit: dBm Actual Value Range: -120~-25 Default Value: -101
InterRatHoUtranGroup	LdSvBasedHoUtranB1ThdEcn0	ADD INTERRATHOUTRANGROUP MOD INTERRATHOUTRANGROUP LST INTERRATHOUTRANGROUP	LOFD-001043 TDLOFD-001022	Service based inter-RAT handover to UTRAN SRVCC to UTRAN	Meaning: Indicates the Ec/N0 threshold for event B1 related to load- or service-based inter-RAT handover to UTRAN. This parameter specifies the requirement for Ec/N0 of the target UTRAN cell. When the measurement value exceeds this threshold, a measurement report may be sent. For a cell with large signal fading variance, set this parameter to a large value to prevent unnecessary handovers. For a cell with small signal fading variance, set this parameter to a small value to ensure timely handovers. GUI Value Range: -48~0 Unit: 0.5dB Actual Value Range: -24~0 Default Value: -18

Copyright © Huawei Technologies Co., Ltd.

MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
CellAlgoSwitch	MlbHoMode	MOD CELLALGOSWITCH LST CELLALGOSWITCH	LOFD-001032 / TDLOFD-001032 LOFD-001044 / TD LOFD-001044 LOFD-001045 / TDLOFD-001045	Intra-LTE Load Balancing Inter-RAT Load Sharing to UTRAN Inter-RAT Load Sharing to GERAN	Meaning: Indicates whether to enable or disable handovers triggered by mobility load balancing (MLB), based on which the eNodeB determines handover policies. If the InterFreqMlbBlindHo check box under this parameter is selected, the eNodeB performs a blind handover to an intra-RAT inter-frequency neighboring cell for load balancing. Otherwise, the eNodeB does not perform the blind handover. If the InterFreqMlbRedirect check box under this parameter is selected, the eNodeB performs a load-based redirection to an intra-RAT inter-frequency neighboring cell. Otherwise, the eNodeB does not perform the redirection. This check box is dedicated to LTE TDD cells. If the InterRatMlbRedirect check box under this parameter is selected, the eNodeB performs a redirection from an E-UTRAN cell to a UTRAN or GERAN cell. Otherwise, the eNodeB does not perform the redirection. This check box is dedicated to LTE TDD cells. If the InterRatMlbBlindHo check box under this parameter is selected, the eNodeB performs a blind handover from an E-UTRAN cell to a UTRAN or GERAN cell for load balancing. Otherwise, the eNodeB does not perform the blind handover. GUI Value Range: InterFreqMlbBlind-Ho(InterFreqMlbBlindHo), InterRatMlbBlind-Ho(InterRatMlbBlindHo), InterFreqMlbRedir-ect(InterFreqMlbRedirect), InterRatMlbRedir-ect(InterRatMlbRedirect) Unit: None Actual Value Range: InterFreqMlbBlindHo, InterRatMlbBlindHo, InterFreqMlbRedirect, InterRatMlbRedirect Default Value: InterFreqMlbBlindHo:Off, InterRatMlbBlindHo:Off, InterFreqMlbRedirect:Off, InterRatMlbRedirect:Off

Copyright © Huawei Technologies Co., Ltd.

MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
CellMLB	InitValidPeriod	MOD CELLMLB LST CELLMLB	LOFD-001044 / TDLOFD-001044	Inter-RAT Load Sharing to UTRAN	Meaning: Indicates the standard valid duration for load sharing with universal terrestrial radio access network (UTRAN) for UEs in idle mode. The actual duration increases on the basis of a standard valid duration with the increase of the number of uplink-synchronized UEs. GUI Value Range: 1~30 Unit: s Actual Value Range: 1~30 Default Value: 10
GeranNcell	NoHoFlag	ADD GERANNCELL MOD GERANNCELL LST GERANNCELL	LOFD-001020 / TDLOFD-001020	PS Inter-RAT Mobility between E-UTRAN and GERAN	Meaning: Indicates whether handovers of UEs to the neighboring cell are prohibited. GUI Value Range: PERMIT_HO_ENUM(Permit Ho), FORBID_HO_ENUM(Forbid Ho) Unit: None Actual Value Range: PERMIT_HO_ENUM, FORBID_HO_ENUM Default Value: PERMIT_HO_ENUM(Permit Ho)
InterRatHoGeranGroup	LdSvBasedHoGeranB1Thd	ADD INTERRATHOGERANGROUP MOD INTERRATHOGERANGROUP LST INTERRATHOGERANGROUP	LOFD-001046 / TDLOFD-001046	Service based inter-RAT handover to GERAN	Meaning: Indicates the RSSI threshold for event B1 related to load- or service-based inter-RAT handover to GERAN. When the measured RSSI value exceeds this threshold, a measurement report will be sent. GUI Value Range: -110~-48 Unit: dBm Actual Value Range: -110~-48 Default Value: -98

Copyright © Huawei Technologies Co., Ltd.

MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
SpidCfg	Spid	ADD SPIDCFG LST SPIDCFG MOD SPIDCFG RMV SPIDCFG	LOFD-00105401 / TDLOFD-00105401 LOFD-001059 / TDLOFD-001059	Camp & Handover Based on SPID UL Pre-allocation Based on SPID	Meaning: Indicates the subscriber profile ID (SPID). GUI Value Range: 1~256 Unit: None Actual Value Range: 1~256 Default Value: None
SpidCfg	RatFreqPriorityInd	ADD SPIDCFG MOD SPIDCFG LST SPIDCFG	LOFD-00105401 / TDLOFD-00105401	Camp & Handover Based on SPID	Meaning: Indicates whether to set the priority for the intra-RAT frequency group. If this parameter is set to CFG(CFG), UEs select the target frequency based on the priority setting. If this parameter is set to NOT_CFG(NOT_CFG), UEs do not select the target frequency based on the priority setting. GUI Value Range: NOT_CFG(NOT_CFG), CFG(CFG) Unit: None Actual Value Range: NOT_CFG, CFG Default Value: NOT_CFG(NOT_CFG)
SpidCfg	RatFreqPriorityGroupId	ADD SPIDCFG MOD SPIDCFG LST SPIDCFG	LOFD-00105401 / TDLOFD-00105401	Camp & Handover Based on SPID	Meaning: Indicates the ID of the intra-RAT frequency priority group. GUI Value Range: 0~255 Unit: None Actual Value Range: 0~255 Default Value: 0

Copyright © Huawei Technologies Co., Ltd.

MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
CellMLB	InterFreqLoadEvalPrd	MOD CELLMLB LST CELLMLB	LOFD-070215 / TDLOFD-070215	Intra-LTE User Number Load Balancing	Meaning: Indicates the evaluation period for determining whether to trigger inter-frequency load balancing. If the serving cell remains in the heavy load state within an evaluation period, inter-frequency load balancing is triggered and certain UEs are transferred for load balancing. If the UtranMlbSwitch or GeranMlbSwitch option of the MlbAlgoSwitch parameter is selected, the InterFreqMlbSwitch or InterFreqBlindMlbSwitch option of the MlbAlgoSwitch parameter is selected, and the FreqSelectStrategy parameter is set to PRIORITYBASED(PriorityBased), the values of the InterRatLoadEvalPrd and InterFreqLoadEvalPrd parameters must be the same. If the two parameters are set to different values, the evaluation period specified by the InterRatLoadEvalPrd parameter is adapted when both cell-level inter-frequency load balancing and inter-RAT load sharing are triggered at the same time. GUI Value Range: 5~30 Unit: s Actual Value Range: 5~30 Default Value: 30

Copyright © Huawei Technologies Co., Ltd.

10 Counters

Table 10-1 Counters

Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526728443	L.RRCRel.DedicatedPri.WCDMA.High	Number of times that the WCDMA network is configured as the highest frequency priority in an IMMCI IE	Multi-mode: None GSM: None UMTS: None LTE: LOFD-00105401 TDLOFD-00105401 LOFD-001044 TDLOFD-001044 LOFD-001112 TDLOFD-001112	Camp & Handover Based on SPID Camp & Handover Based on SPID Inter-RAT Load Sharing to UTRAN Inter-RAT Load Sharing to UTRAN MOCN Flexible Priority Based Camping MOCN Flexible Priority Based Camping
1526729878	L.RRCRel.load.DedicatedPri.WCDMA.High	Number of times that the WCDMA network is configured as the highest frequency priority in an IMMCI IE Triggered Because of High Load	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001044 TDLOFD-001044	Inter-RAT Load Sharing to UTRAN Inter-RAT Load Sharing to UTRAN

Copyright © Huawei Technologies Co., Ltd.

Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526728949	L.RIM.Load.E2W.Req	Number of times the eNodeB sends a load information request to a WCDMA network	Multi-mode: MRFD-090211 GSM: None UMTS: None LTE: LOFD-001044 TDLOFD-001044	eCoordinator based RIM process optimization Inter-RAT Load Sharing to UTRAN Inter-RAT Load Sharing to UTRAN
1526728950	L.RIM.Load.E2W.Resp	Number of times the eNodeB receives a load information response from a WCDMA network	Multi-mode: MRFD-090211 GSM: None UMTS: None LTE: LOFD-001044 TDLOFD-001044	eCoordinator based RIM process optimization Inter-RAT Load Sharing to UTRAN Inter-RAT Load Sharing to UTRAN
1526728951	L.RIM.Load.E2W.Update	Number of times the eNodeB receives a load information update from a WCDMA network	Multi-mode: MRFD-090211 GSM: None UMTS: None LTE: LOFD-001044 TDLOFD-001044	eCoordinator based RIM process optimization Inter-RAT Load Sharing to UTRAN Inter-RAT Load Sharing to UTRAN
1526728529	L.IRATHO.E2W.Load.PrepAttOut	Number of inter-RAT handover preparation attempts from E-UTRAN to WCDMA network triggered because of high load	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001044 TDLOFD-001044 LOFD-001022 TDLOFD-001022	Inter-RAT Load Sharing to UTRAN Inter-RAT Load Sharing to UTRAN SRVCC to UTRAN SRVCC to UTRAN
1526728530	L.IRATHO.E2W.Load.ExecAttOut	Number of inter-RAT handover execution attempts from E-UTRAN to WCDMA network triggered because of high load	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001044 TDLOFD-001044 LOFD-001022 TDLOFD-001022	Inter-RAT Load Sharing to UTRAN Inter-RAT Load Sharing to UTRAN SRVCC to UTRAN SRVCC to UTRAN

Copyright © Huawei Technologies Co., Ltd.

Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526728531	L.IRATHO.E2W.Load.ExecSuccOut	Number of successful inter-RAT handover executions from E-UTRAN to WCDMA network triggered because of high load	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001044 TDLOFD-001044 LOFD-001022 TDLOFD-001022	Inter-RAT Load Sharing to UTRAN Inter-RAT Load Sharing to UTRAN SRVCC to UTRAN SRVCC to UTRAN
1526728532	L.IRATHO.E2G.Load.PrepAttOut	Number of inter-RAT handover preparation attempts from E-UTRAN to GERAN triggered because of high load	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001045 TDLOFD-001045 LOFD-001023 TDLOFD-001023	Inter-RAT Load Sharing to GERAN Inter-RAT Load Sharing to GERAN SRVCC to GERAN SRVCC to GERAN
1526728533	L.IRATHO.E2G.Load.ExecAttOut	Number of inter-RAT handover execution attempts from E-UTRAN to GERAN triggered because of high load	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001045 TDLOFD-001045 LOFD-001023 TDLOFD-001023	Inter-RAT Load Sharing to GERAN Inter-RAT Load Sharing to GERAN SRVCC to GERAN SRVCC to GERAN
1526728534	L.IRATHO.E2G.Load.ExecSuccOut	Number of successful inter-RAT handover executions from E-UTRAN to GERAN triggered because of high load	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001045 TDLOFD-001045 LOFD-001023 TDLOFD-001023	Inter-RAT Load Sharing to GERAN Inter-RAT Load Sharing to GERAN SRVCC to GERAN SRVCC to GERAN
1526728535	L.RRCRedirection.E2W.Load	Number of redirections from E-UTRAN to WCDMA network triggered because of high load	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001044 TDLOFD-001044	Inter-RAT Load Sharing to UTRAN Inter-RAT Load Sharing to UTRAN

Copyright © Huawei Technologies Co., Ltd.

Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526728536	L.RRCRedirection.E2G.Load	Number of redirections from E-UTRAN to GERAN triggered because of high load	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001045 TDLOFD-001045	Inter-RAT Load Sharing to GERAN Inter-RAT Load Sharing to GERAN

Copyright © Huawei Technologies Co., Ltd.

11 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

Copyright © Huawei Technologies Co., Ltd.

12 Reference Documents

1. 3GPP TS 36.314: "Evolved Universal Terrestrial Radio Access Network (E-UTRAN); S1 Application Protocol (S1AP)"
2. 3GPP TS 36.423: "Evolved Universal Terrestrial Radio Access Network (E-UTRAN); X2 Application Protocol (X2AP)"
3. 3GPP TS 36.331: "Evolved Universal Terrestrial Radio Access (E-UTRA); Radio Resource Control (RRC); Protocol specification"
4. 3GPP TS 48.018: "General Packet Radio Service (GPRS); Base Station System (BSS) - Serving GPRS Support Node (SGSN); BSS GPRS protocol (BSSGP)"
5. *CS Fallback Feature Parameter Description*
6. *Inter-RAT Mobility Management in Connected Mode Feature Parameter Description*

Copyright © Huawei Technologies Co., Ltd.